MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3660 LOW GRADE ACID WASH/ELUTION/CARBON REGEN.
DESIGN OF PLATFORM FOUNDATIONS

Document Number: **MD-216-3660-EG-CV-CAL-2002**
Revision: **D01**
Revision Date: **27.11.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	EE13	CED01	36603	OU100705936	2
L&T Document Number:	O19045	450	C	K	24449	002	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	10.11.2020	FIRST ISSUE	TEC	AVG	SJM	
C01	08.01.2021	ISSUED FOR CONSTRUCTION	TEC	AVG	SJM	
D01	27.11.2023	AS BUILT	TEC	AVG	SJM	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Regen.	:	Regeneration

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1. INTENT

The purpose of this document is to analysis and design of foundations in **LOW GRADE ACID WASH/ELUTION/CARBON REGEN** area. In this document design & analysis of Platform, Column & Pump foundation have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

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3. SCOPE

The Tank Foundation design has been carried out using excel sheets. Design data of tanks are taken from document no. MD-216-3660-EG-ME-GAD-1004-A01, MD-216-3660-EG-ME-GAD-1034-A01, MD-216-3660-EG-ME-GAD-1035-A01, MD-216-3660-EG-ME-GAD-1037-A01, MD-216-3660-EG-ME-GAD-1038-A01. Tanks for which foundation design covered in this document are 3660-TNK-001, 3660-TNK-002, 3660-TNK-004, 3660-TNK-005 & 3660-TNK-010.

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.2 Loading

A. Primary Load for Design of Footing F3:

1 : DL1_PUMP_OPERATING	8 : LL1
2 : DL2_PUMP_EMPTY	9 : LL2
3 : DL3_STRUCTURE	10 : LL3
4 : WL1	11 : EL1
5 : WL2	12 : EL2
6 : WL3	13 : SL1_EMP(+X)
7 : WL4	14 : SL2_EMP(+Z)

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B. Primary Load for Design of Platform Foundation:

1 : DEAD LOAD
2 : LIVE-1 LOAD
3 : LIVE-2 LOAD
4 : LIVE-3 LOAD
5 : WIND LOAD (+X DIR)
6 : WIND LOAD (-X DIR)
7 : WIND LOAD(+Z DIR)
8 : WIND LOAD(-Z DIR)
11 : DL
12 : LL1
13 : LL2
14 : LL3
15 : WL1
16 : WL2
17 : WL3
18 : WL4

Basic load cases kept same as used in superstructure design.

4.3 Load combination

4.3.1 For design of footing F3

A. Allowable Stress Design.

202 : 1 (DL1 + DL3) + 1 (LL1+LL2+LL3)
203 : 1 (DL+ DL3) + 1 WL1
204 : 1 (DL+ DL3) + 1 WL2
205 : 1 (DL+ DL3) + 1 WL3
206 : 1 (DL+ DL3) + 1 WL4
207 : 1 (DL1+DL3) + 0.7 EL1
208 : 1 (DL1+DL3) - 0.7 EL1
209 : 1 (DL1+DL3) + 0.7 EL2
210 : 1 (DL1+DL3) - 0.7 EL2
211 : 0.6 (DL2+DL3) + 1 WL1
212 : 0.6 (DL2+DL3) + 1 WL2
213 : 0.6 (DL2+DL3) + 1 WL3
214 : 0.6 (DL2+DL3) + 1 WL4
215 : 0.6 (DL2+DL3) + 0.7 SL1_EMP(+X)
216 : 0.6 (DL2+DL3) - 0.7 SL1_EMP(+X)
217 : 0.6 (DL2+DL3) + 0.7 SL1_EMP(+Z)
218 : 0.6 (DL2+DL3) - 0.7 SL1_EMP(+Z)

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219 : 1 (DL1+DL3)+ 0.75 WL1 + 0.75(LL1+LL2+LL3)
220 : 1 (DL1+DL3)+ 0.75 WL2 + 0.75(LL1+LL2+LL3)
221 : 1 (DL1+DL3)+ 0.75 WL3 + 0.75(LL1+LL2+LL3)
222 : 1 (DL1+DL3)+ 0.75 WL4 + 0.75(LL1+LL2+LL3)
223 : 1 (DL1 + DL3) + 0.525 EL1
224 : 1 (DL1 + DL3) - 0.525 EL1
225 : 1 (DL1 + DL3) + 0.525 EL2
226 : 1 (DL1 + DL3) - 0.525 EL2

B. Ultimate Strength Design.

101 : 1.4 (DL1+DL3)
102 : 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
103 : 1.2 (DL1+DL3)+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
104 : 1.2 (DL1+DL3)+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)
105 : 1.2 (DL1+DL3)+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)
106 : 1.2 (DL1+DL3)+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)
107 : 1.2 (DL1+DL3) + 1.0 EL1
108 : 1.2 (DL1+DL3) - 1.0 EL1
109 : 1.2 (DL1+DL3) + 1.0 EL2
110 : 1.2 (DL1+DL3) - 1.0 EL2
111 : 0.9 (DL2+DL3) + 1.6 WL1
112 : 0.9 (DL2+DL3) + 1.6 WL2
113 : 0.9 (DL2+DL3) + 1.6 WL3
114 : 0.9 (DL2+DL3) + 1.6 WL4
115 : 0.9 (DL2+DL3) + 1.0 SL1_EMP(+X)
116 : 0.9 (DL2+DL3) - 1.0 SL1_EMP(+X)
117 : 0.9 (DL2+DL3) + 1.0 SL1_EMP(+Z)
118 : 0.9 (DL2+DL3) - 1.0 SL1_EMP(+Z)
119 : 1.2 (DL1+DL3) + 0.8 WL1
120 : 1.2 (DL1+DL3) + 0.8 WL2
121 : 1.2 (DL1+DL3) + 0.8 WL3
122 : 1.2 (DL1+DL3) + 0.8 WL4

4.3.2 For design of footing F6 to F9

A. Allowable Stress Design.

201 : 1. DL
202 : 1 DL + 1 (LL1+LL2+LL3)
203 : 1 DL+ 1 WL1
204 : 1 DL+ 1 WL2
205 : 1 DL+ 1 WL3
206 : 1 DL+ 1 WL4
207 : 0.6 DL + 1 WL1
208 : 0.6 DL + 1 WL2
209 : 0.6 DL + 1 WL3
210 : 0.6 DL + 1 WL4
211 : 1 DL+ 0.75 WL1+ 0.75 (LL1+LL2+LL3)
212 : 1 DL+ 0.75 WL2+ 0.75 (LL1+LL2+LL3)
213 : 1 DL+ 0.75 WL3+ 0.75 (LL1+LL2+LL3)
214 : 1 DL+ 0.75 WL4+ 0.75 (LL1+LL2+LL3)

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B. Ultimate Strength Design.

101 : 1.4 DL
102 : 1.2 (DL) + 1.6 (LL1+LL2+LL3)
103 : 1.2 DL+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
104 : 1.2 DL+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)
105 : 1.2 DL+ 1.6 WL3 + 0.5 (LL1+LL2+LL3)
106 : 1.2 DL+ 1.6 WL4 + 0.5 (LL1+LL2+LL3)
107 : 0.9 DL + 1.6 WL1
108 : 0.9 DL + 1.6 WL2
109 : 0.9 DL + 1.6 WL3
110 : 0.9 DL + 1.6 WL4
111 : 1.2 DL + 0.8 WL1
112 : 1.2 DL + 0.8 WL2
113 : 1.2 DL + 0.8 WL3
114 : 1.2 DL + 0.8 WL4

4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m2 at 1.5m depth below EGL (RL 912.200) refer Table 5 - BH-11 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

The Vertical Modulus of Subgrade Reaction has been considered as 40500 kN/m^2/m at 1.5m depth below EGL (RL 912.500) refer Table 5 - BH-11 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

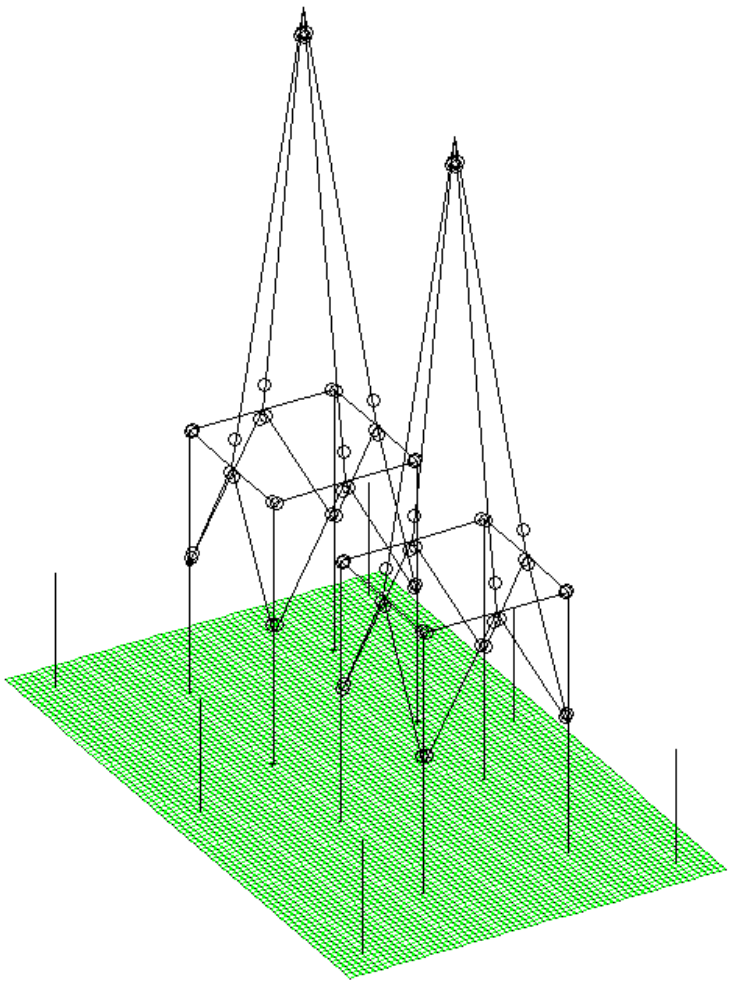
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5.1 Design of Foundation F3

This is a combined footing hosting the supporting structure for equipment 3660-COL-001 & 002 and 6 number of columns supporting platform above.

In the absence of accurate equipment loading, STAAD model created to represent equipment COG (by rigid members) and calculated loads are applied at CG location to estimate forces at equipment base. Platform column reactions are obtained from the STAAD file created to perform superstructure analysis and design.

Due to close proximity of column a combined mat is provided in entire area. See snap below:



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Wind & Seismic forces are calculated manually based on column geometry. See below the calculation

WIND LOAD CALCULATION

Basic Wind Speed (Vs) :	45m/s	=162Km/h
Velocity pressure coeff (Kd) :	1.00	
Exposure Category :	C	
Topography factor (Kzt) :	1	
Exposure coefficient (Kz) :	1	(Refer table 7.2-2 of SBC-301)
Tank Height (H) :	12.05	m
Tank Diameter (D) :	2	m

Velocity presure (qz) = $q_z = 0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ (kN/m^2); V in km/h (Ref eq. 7.2-12 of SBC-301) = 1.43kN/m2

Force coefficient, Cf = 0.8 (Refer table 7.2-15 of SBC-301)

Thus, Base shear due to wind = 1.43x0.8x12.05x2= 27.57 kN

Moment at base due to wind = 27.5704x12.05x0.5= 166.11 kN.m

SEISMIC LOAD CALCULATION

Spectral response accelaration, S1 =	0.10g
Spectral response accelaration, Ss =	0.02g
Site Class :	B
Site coefficient, Fa =	1
Site coefficient, Fv =	1
Design Spectral response accelaration, Sd1 =	0.013g
Design Spectral response accelaration, Sds =	0.067g
Time Period (TL) :	4 sec
Liquid height, H1 =	2.5 m

The seismic base shear is the combination of the impulsive and convective components:

V = Vi + Vc

$$V_i = \frac{S_{ai} W_i}{(R/I)} \qquad V_c = \frac{S_{ac} I}{1.5} W_c$$

Conservatively considering Sai = Sds, and R = 1, I = 1.15

Hence, Vi = 28.51 kN

Natural time period for first convective mode of sloshing :

$$T_c = 2\pi \sqrt{\frac{D}{3.68\, g\, \tan h\left(\frac{3.68H}{D}\right)}} = 2.3088 \text{ sec}$$

For $T_c > T_L$:

$$S_{ac} = \frac{1.5\, S_{D1} T_L}{T_c^2} = 0.01463$$

For $T_c \le T_L$:

$$S_{ac} = \frac{1.5\, S_{D1}}{T_c} = 0.008446$$

Thus, Vc = 2.15 kN

Hence, seismic base shear, V = 30.66 kN

Moment at base due to seismic = 30.66x12.05x0.5= 184.73 kN.m

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Maximum pressure appearing at foundation base is less than the allowable bearing capacity of soil. See below the maximum pressure at footing base:

	Node	L/C	Horizontal Fx N/mm2	Vertical Fy N/mm2	Horizontal Fz N/mm2
Max Px	18	201	0	0.036	0
Min Px	18	201	0	0.036	0
Max Py	49	220	0	0.13	0
Min Py	48	212	0	-0.016	0
Max Pz	18	201	0	0.036	0
Min Pz	18	201	0	0.036	0

Gross Maximum Base pressure, p_{max} = 130.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2 (Bore hole 27)
Depth of soil above footing = 0.45 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 458.1 KN/m^2
> 130.0 KN/m^2
Hence OK

Footing parameter consider in design is mentioned below:

Size of fdn. raft:	6.00m	9.50m
Thickness of raft:	0.75	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above G	0.55	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Note: Overturning and sliding check not performed for this raft. This is considered as OK by engineering judgement.

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Design of Foundation Raft F3 :

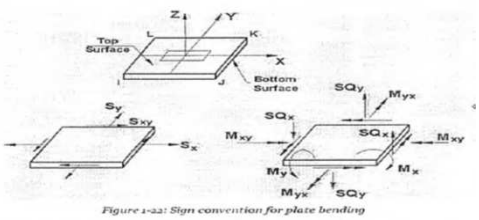
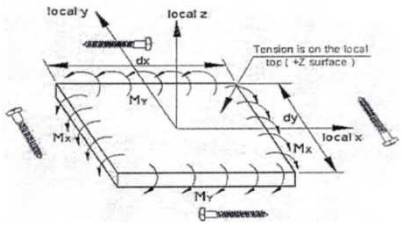


Figure 1-2a: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; : FDN_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	2706	0.371	0.113	0	0	0	-19.01	233.663	-15.72	-34.729	249.383	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Min Qx	2711	-1.006	0.171	0	0	0	128.64	190.563	0	128.642	190.563	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Max Qy	2589	-0.266	0.867	0	0	0	106.47	186.332	-4.141	110.61	190.473	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Min Qy	2889	-0.24	-0.765	0	0	0	112.56	113.169	-2.846	115.402	116.015	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Max Sx	64	0.009	-0.005	0	0	0	0.822	0.418	-0.45	1.272	0.868	101 1.4 (DL1+DL3)
Min Sx	64	0.009	-0.005	0	0	0	0.822	0.418	-0.45	1.272	0.868	101 1.4 (DL1+DL3)
Max Sy	64	0.009	-0.005	0	0	0	0.822	0.418	-0.45	1.272	0.868	101 1.4 (DL1+DL3)
Min Sy	64	0.009	-0.005	0	0	0	0.822	0.418	-0.45	1.272	0.868	101 1.4 (DL1+DL3)
Max Sxy	64	0.009	-0.005	0	0	0	0.822	0.418	-0.45	1.272	0.868	101 1.4 (DL1+DL3)
Min Sxy	64	0.009	-0.005	0	0	0	0.822	0.418	-0.45	1.272	0.868	101 1.4 (DL1+DL3)
Max Mx	2771	-0.716	-0.133	0	0	0	282.09	56.787	38.53	320.623	95.317	103 1.2 (DL1+DL3)+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
Min Mx	2816	-0.242	0.029	0	0	0	-265.9	136.189	51.56	-317.497	187.749	103 1.2 (DL1+DL3)+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
Max My	2706	0.371	0.113	0	0	0	-19.01	233.663	-15.72	-34.729	249.383	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Min My	4266	-0.028	0.045	0	0	0	-7.012	-169.23	17.745	-24.757	-186.975	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Max Mxy	2876	-0.159	0.081	0	0	0	-192.7	98.34	104.457	-297.205	202.797	103 1.2 (DL1+DL3)+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
Min Mxy	2646	-0.189	-0.103	0	0	0	-58.87	50.106	-87.439	-146.304	137.545	103 1.2 (DL1+DL3)+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)

PLATE CENTER STRESS ; FDN_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	5752	0.306	0.009	0	0	0	-30.01	-0.514	-8.803	-38.808	-9.317	105 1.2 (DL1+DL3)+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)
Min Qx	5715	-0.633	-0.001	0	0	0	-45.6	0.332	0	-45.603	0.332	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Max Qy	2343	0.017	0.494	0	0	0	-0.701	50.265	-10.075	-10.776	60.34	103 1.2 (DL1+DL3)+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
Min Qy	3183	0.016	-0.609	0	0	0	-0.865	73.185	14.849	-15.714	88.034	103 1.2 (DL1+DL3)+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
Max Sx	75	-0.099	0.002	0	0	0	2.273	0.125	-2.02	4.293	2.145	101 1.4 (DL1+DL3)
Min Sx	75	-0.099	0.002	0	0	0	2.273	0.125	-2.02	4.293	2.145	101 1.4 (DL1+DL3)
Max Sy	75	-0.099	0.002	0	0	0	2.273	0.125	-2.02	4.293	2.145	101 1.4 (DL1+DL3)
Min Sy	75	-0.099	0.002	0	0	0	2.273	0.125	-2.02	4.293	2.145	101 1.4 (DL1+DL3)
Max Sxy	75	-0.099	0.002	0	0	0	2.273	0.125	-2.02	4.293	2.145	101 1.4 (DL1+DL3)
Min Sxy	75	-0.099	0.002	0	0	0	2.273	0.125	-2.02	4.293	2.145	101 1.4 (DL1+DL3)
Max Mx	2715	-0.26	0.016	0	0	0	80.844	65.465	5.476	86.32	70.941	103 1.2 (DL1+DL3)+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
Min Mx	5614	-0.051	0.027	0	0	0	-187	-6.429	16.968	-203.925	-23.397	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Max My	2715	-0.36	0.002	0	0	0	-23.86	109.778	6.342	-30.2	116.12	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Min My	4266	-0.028	0.045	0	0	0	-7.012	-169.23	17.745	-24.757	-186.975	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Max Mxy	5235	-0.162	0.127	0	0	0	-52.21	-33.935	84.325	-136.53	-118.26	102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
Min Mxy	5392	0.009	-0.007	0	0	0	-33.15	-22.895	-36.959	-70.113	-59.854	105 1.2 (DL1+DL3)+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		750	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

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Reinforcement Along X:

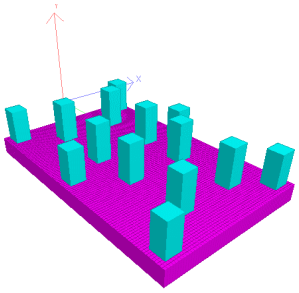
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	320.62	top face	50	690	320.62	0.90	0.0022	1498.1	1350	20	209.79	200	1572	22.16	27.7	0.072	tension control
Min Mx	-317.50	bottom face	50	690	317.5	0.89	0.0021	1483.3		20	211.89	200	1571.5	22.16	27.7	0.072	tension control
Max Mxy	-297.21	bottom face	50	690	297.21	0.83	0.0020	1387.1		20	226.58	200	1572	22.16	27.7	0.072	tension control
Min Mxy	-146.30	bottom face	50	690	146.3	0.41	0.0010	677.8		20	232.8	200	1572	22.16	27.7	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	Max Qx	0.306		229.5	38.808	0.002	0.69	762.147	679.851	1183.81	679.85	2531	> 229.5	ok	509.9	φVc>Vu	not required
	Min Qx	0.633		474.75	45.603			845.039			679.85	2531	> 474.75	ok	509.9	φVc>Vu	not required

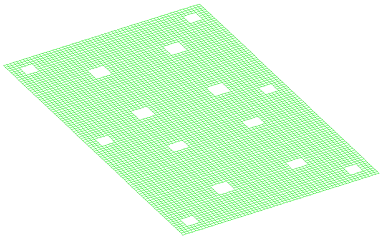
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	249.38	top face	50	690	249.38	0.70	0.0017	1161.2	1350	20	232.8	200	1571.5	22.16	27.7	0.072	tension control
Min My	-186.98	bottom face	50	690	186.98	0.52	0.0013	868.0		20	232.8	200	1571.5	22.16	27.7	0.072	tension control
Max Mxy	202.80	top face	50	690	202.8	0.57	0.0014	942.1		20	232.8	200	1571.5	22.16	27.7	0.072	tension control
Min Mxy	137.55	top face	50	690	137.55	0.39	0.0009	637.0		20	232.8	200	1571.5	22.16	27.7	0.072	tension control

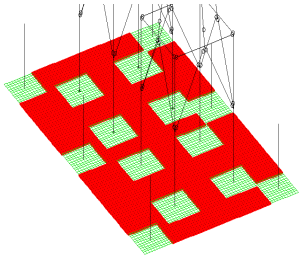
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	Max Qy	0.494		370.5	60.34	0.002	0.69	fc'766.3	fc'679.9	fc'1184	679.85	712	> 370.5	ok	509.9	φVc>Vu	not required
	Min Qy	0.609		456.75	88.034			748.775			679.85	712	> 456.75	ok	509.9	φVc>Vu	not required



3D Rendered View of foundation



Highlighted plate for moment (at column face)



Highlighted plate for shear (at d distance from column face)

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
- At Bottom **20ϕT** **200** c/c
- At Top **20ϕT** **200** c/c

Along Z Direction :
- At Bottom **20ϕT** **200** c/c
- At Top **20ϕT** **200** c/c

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5.2 Design of Foundation F6

	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.50 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.20 m	
U/S level Base Plate :	0.55 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

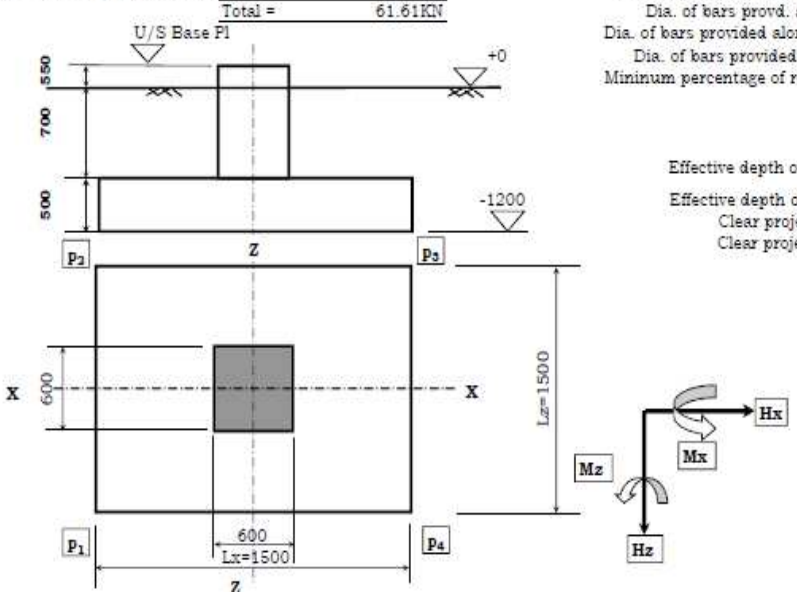
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.5*1.5*0.5*24)+(0.6*0.6*1.25*24) =	37.80KN
Weight of soil = ((1.5*1.5)-(0.6*0.6))*0.7*18 =	23.81KN
Total =	61.61KN

Design Input:

Characteristic strength of concrete, f'$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%



Effective depth of slab along X direction = 442mm = 0.442m

Effective depth of slab along Z direction = 426mm = 0.426m

Clear projection along Z direction = 0.45m

Clear projection along X direction = 0.45m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201 1. DL		9.8	-0.8	0.0	0.8	0.0	70.89	-0.81	-0.55	0.81	0.55
202 1 DL + 1 (LL1+LL2+LL3)		42.7	-2.8	0.0	-1.0	0.0	104.81	-2.82	-4.06	-1.04	-1.82
203 1 DL+ 1 WL1		22.4	-7.9	0.0	0.0	0.0	84.04	-7.94	-13.89	-0.03	-0.06
204 1 DL+ 1 WL2		-3.9	7.8	0.0	0.7	0.0	57.74	7.81	12.79	0.66	1.16
205 1 DL+ 1 WL3		9.0	-0.8	0.0	-7.6	0.0	70.62	-0.84	-0.60	-7.64	-13.38
206 1 DL+ 1 WL4		9.6	-0.8	0.0	8.8	0.0	71.16	-0.28	-0.49	8.27	14.47
207 0.6 DL + 1 WL1		18.7	-7.8	0.0	-0.2	0.0	80.88	-7.81	-13.67	-0.16	-0.28
208 0.6 DL + 1 WL2		-7.6	7.4	0.0	0.5	0.0	54.08	7.44	13.01	0.54	0.94
209 0.6 DL + 1 WL3		5.8	-0.2	0.0	-7.8	0.0	66.91	-0.22	-0.38	-7.77	-13.59
210 0.6 DL + 1 WL4		5.8	-0.2	0.0	8.1	0.0	67.45	-0.16	-0.27	8.15	14.25
211 1 DL+ 0.75 WL1+ 0.75 (LL1+LL2+LL3)		44.2	-7.5	0.0	-1.0	0.0	105.82	-7.54	-13.19	-0.96	-1.69
212 1 DL+ 0.75 WL2+ 0.75 (LL1+LL2+LL3)		24.5	3.9	0.0	-0.4	0.0	86.09	3.90	6.82	-0.44	-0.77
213 1 DL+ 0.75 WL3+ 0.75 (LL1+LL2+LL3)		34.1	-1.8	0.0	-6.7	0.0	95.76	-1.84	-3.23	-6.67	-11.67
214 1 DL+ 0.75 WL4+ 0.75 (LL1+LL2+LL3)		34.5	-1.8	0.0	5.8	0.0	96.16	-1.80	-3.14	5.27	9.22
Ultimate Conditions											
101 1.4 DL		13.0	-0.4	0.0	0.4	0.0	74.61	-0.44	-0.77	0.44	0.77
102 1.2 (DL) + 1.6 (LL1+LL2+LL3)		64.6	-3.6	0.0	-1.8	0.0	126.22	-3.59	-6.28	-1.79	-3.13
103 1.2 DL+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)		48.9	-13.6	0.0	-0.9	0.0	110.50	-13.58	-23.76	-0.86	-1.50
104 1.2 DL+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)		6.8	10.8	0.0	0.8	0.0	68.42	10.82	18.98	0.26	0.45
105 1.2 DL+ 1.6 WL3 + 0.5 (LL1+LL2+LL3)		27.4	-1.4	0.0	-10.0	0.0	89.08	-1.43	-2.50	-10.03	-22.80
106 1.2 DL+ 1.6 WL4 + 0.5 (LL1+LL2+LL3)		28.8	-1.8	0.0	12.4	0.0	89.89	-1.83	-2.88	12.43	21.75
107 0.9 DL + 1.6 WL1		29.4	-12.5	0.0	-0.8	0.0	91.01	-12.48	-21.84	-0.28	-0.48
108 0.9 DL + 1.6 WL2		-12.7	11.9	0.0	0.8	0.0	48.92	11.92	20.85	0.84	1.47
109 0.9 DL + 1.6 WL3		7.9	-0.8	0.0	-12.4	0.0	69.54	-0.83	-0.58	-12.45	-21.78
110 0.9 DL + 1.6 WL4		8.8	-0.2	0.0	13.0	0.0	70.39	-0.28	-0.41	13.01	22.77
111 1.2 DL + 0.8 WL1		21.7	-6.5	0.0	0.1	0.0	83.27	-6.48	-11.33	0.10	0.17
112 1.2 DL + 0.8 WL2		0.6	5.7	0.0	0.7	0.0	62.28	5.72	10.02	0.66	1.15
113 1.2 DL + 0.8 WL3		10.9	-0.4	0.0	-6.0	0.0	72.58	-0.40	-0.70	-5.99	-10.48
114 1.2 DL + 0.8 WL4		11.4	-0.4	0.0	6.7	0.0	72.96	-0.85	-0.61	6.74	11.80

Check for overturning and sliding:										
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL		70.9	-0.8	-0.5	0.8	0.5	97.1	96.8	90.6	90.8
202 1 DL + 1 (LL1+LL2+LL3)		104.8	-2.8	-4.1	-1.0	-1.8	19.8	48.0	18.0	40.1
203 1 DL+ 1 WL1		84.0	-7.9	-13.9	0.0	-0.1	4.5	1059.4	4.2	988.7
204 1 DL+ 1 WL2		57.7	7.8	12.8	0.7	1.2	8.4	37.4	8.2	34.9
205 1 DL+ 1 WL3		70.6	-0.8	-0.6	-7.6	-13.4	88.0	4.0	82.1	3.7
206 1 DL+ 1 WL4		71.2	-0.8	-0.5	8.8	14.5	108.1	3.7	100.9	3.4
207 0.6 DL + 1 WL1		80.8	-7.8	-13.7	-0.2	-0.8	4.4	215.2	4.1	200.8
208 0.6 DL + 1 WL2		54.0	7.4	13.0	0.5	0.9	8.1	48.1	2.9	40.2
209 0.6 DL + 1 WL3		66.9	-0.2	-0.4	-7.8	-13.6	130.9	3.7	122.2	3.4
210 0.6 DL + 1 WL4		67.4	-0.2	-0.8	8.1	14.8	184.1	3.5	171.8	3.3
211 1 DL+ 0.75 WL1+ 0.75 (LL1+LL2+LL3)		105.8	-7.5	-13.2	-1.0	-1.7	6.0	47.1	5.6	44.0
212 1 DL+ 0.75 WL2+ 0.75 (LL1+LL2+LL3)		86.1	3.9	6.8	-0.4	-0.8	9.5	83.9	8.8	78.3
213 1 DL+ 0.75 WL3+ 0.75 (LL1+LL2+LL3)		95.8	-1.8	-3.2	-6.7	-11.7	22.8	6.2	20.8	5.7
214 1 DL+ 0.75 WL4+ 0.75 (LL1+LL2+LL3)		96.2	-1.8	-3.1	5.8	9.2	22.9	7.8	21.4	7.8
						Min.=	3.1	3.5	2.9	3.3
							OK	OK	OK	OK

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4.9 DESIGN OF FOUNDATION MKD. F6

Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mx/((Lx)^2.Lz)) + (6.Mx/(Lx.(Lz)^2))$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mx/((Lx)^2.Lz)) - (6.Mx/(Lx.(Lz)^2))$

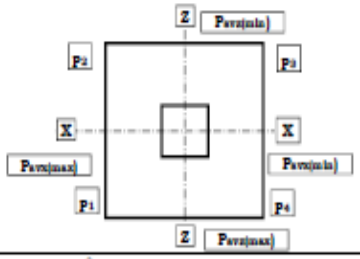
Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m²

Gross SBC of soil for DL, LL= 471.6 kN/m²

For WL/SL case = 562.5 kN/m²

For WL/SL case = 584.1 kN/m²



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/(Lx.Lz)	6.Mz/(Lx)².Lz	6.Mx/Lx.(Lz)²	Base Pressure (kN/m²) P1 (Pmax)	P2	P4	P3 (Pmin)	(Pmax-γh)	Pave(max)	Pave(min)	Pave(max)	Pave(min)	Remarks
201	1. DL	70.89	-0.55	0.55	31.51	0.97	0.98	33.46	31.50	31.51	29.56	11.86	32.48	30.54	32.49	30.53	O.K
202	1 DL + 1 (LL1+LL2+LL3)	104.31	-4.06	-1.82	46.36	7.22	3.24	56.82	50.35	42.37	35.90	35.22	53.59	39.14	49.60	43.13	O.K
203	1 DL+ 1 WL1	84.04	-13.89	-0.06	37.35	24.70	0.11	62.15	61.94	12.76	12.55	40.55	62.05	12.66	37.46	37.25	O.K
204	1 DL+ 1 WL2	57.74	12.79	1.16	25.66	22.75	2.06	50.47	46.35	4.98	0.86	28.87	48.41	2.92	27.73	23.61	O.K
205	1 DL+ 1 WL3	70.62	-0.60	-13.38	31.39	1.07	23.78	56.24	8.68	54.10	6.54	34.64	32.46	30.32	55.17	7.61	O.K
206	1 DL+ 1 WL4	71.16	-0.49	14.47	31.63	0.88	25.73	58.24	6.77	56.48	5.02	36.64	32.51	30.75	57.36	5.90	O.K
207	0.6 DL + 1 WL1	80.33	-13.67	-0.28	35.70	24.30	0.50	60.50	59.51	11.90	10.90	38.90	60.01	11.40	36.20	35.21	O.K
208	0.6 DL + 1 WL2	54.03	13.01	0.94	24.01	23.14	1.67	48.82	45.48	2.55	0.00	27.22	47.15	1.28	25.69	22.74	O.K
209	0.6 DL + 1 WL3	66.91	-0.38	-13.59	29.74	0.68	24.17	54.59	6.25	53.22	4.89	32.99	30.42	29.06	53.91	5.57	O.K
210	0.6 DL + 1 WL4	67.45	-0.27	14.25	29.98	0.49	25.34	55.80	5.12	54.83	4.15	34.20	30.46	29.49	55.32	4.64	O.K
211	1 DL+ 0.75 WL1+ 0.75 (LL1+LL2	105.82	-13.19	-1.69	47.03	23.45	3.00	73.48	67.49	26.58	20.58	51.88	70.49	23.58	50.03	44.04	O.K
212	1 DL+ 0.75 WL2+ 0.75 (LL1+LL2	86.09	6.82	-0.77	38.26	12.13	1.37	51.76	49.03	27.50	24.76	30.16	50.40	26.13	39.63	36.90	O.K
213	1 DL+ 0.75 WL3+ 0.75 (LL1+LL2	95.76	-3.23	-11.67	42.56	5.73	20.75	69.04	27.54	57.57	16.08	47.44	48.29	36.83	63.31	21.81	O.K
214	1 DL+ 0.75 WL4+ 0.75 (LL1+LL2	96.16	-3.14	9.22	42.74	5.59	16.38	64.71	31.94	53.53	20.76	43.11	48.33	37.15	59.12	26.35	O.K
	Ultimate Conditions																
101	1.4 DL	74.61	-0.77	0.77	33.16	1.36	1.37	35.89	33.15	33.16	30.43	14.29	34.52	31.80	34.53	31.79	O.K
102	1.2 (DL) + 1.6 (LL1+LL2+LL3)	126.22	-6.28	-3.13	56.10	11.17	5.57	72.84	61.70	50.50	39.36	51.24	67.27	44.93	61.67	50.53	O.K
103	1.2 DL+ 1.6 WL1 + 0.5 (LL1+LL2	110.50	-23.76	-1.50	49.11	42.25	2.67	94.03	88.69	9.53	4.20	72.43	91.36	6.87	51.78	46.45	O.K
104	1.2 DL+ 1.6 WL2 + 0.5 (LL1+LL2	68.42	18.93	0.45	30.41	33.66	0.80	64.87	63.27	0.00	0.00	43.27	64.07	0.00	32.44	31.64	O.K
105	1.2 DL+ 1.6 WL3 + 0.5 (LL1+LL2	89.03	-2.50	-22.80	39.57	4.45	40.54	84.56	3.47	75.66	0.00	62.96	44.02	37.83	80.11	1.74	O.K
106	1.2 DL+ 1.6 WL4 + 0.5 (LL1+LL2	89.89	-2.33	21.75	39.95	4.14	38.67	82.76	5.42	74.48	0.00	61.16	44.09	37.24	78.62	2.71	O.K
107	0.9 DL + 1.6 WL1	91.01	-21.84	-0.48	40.45	38.83	0.86	80.13	78.42	2.47	0.76	58.53	79.28	1.62	41.30	39.59	O.K
108	0.9 DL + 1.6 WL2	48.92	20.85	1.47	21.74	37.06	2.61	61.43	56.21	0.00	0.00	39.83	58.82	0.00	30.72	28.11	O.K
109	0.9 DL + 1.6 WL3	69.54	-0.58	-21.78	30.90	1.03	38.73	70.66	0.00	68.60	0.00	49.06	35.33	34.30	69.63	0.00	O.K
110	0.9 DL + 1.6 WL4	70.39	-0.41	22.77	31.29	0.72	40.48	72.50	0.00	71.05	0.00	50.90	36.25	35.53	71.78	0.00	O.K
111	1.2 DL + 0.8 WL1	83.27	-11.33	0.17	37.01	20.14	0.30	57.46	56.85	17.17	16.56	35.86	57.16	16.87	37.32	36.71	O.K
112	1.2 DL + 0.8 WL2	62.23	10.02	1.15	27.66	17.81	2.04	47.51	43.42	11.89	7.81	25.91	45.47	9.85	29.70	25.62	O.K
113	1.2 DL + 0.8 WL3	72.53	-0.70	-10.48	32.24	1.24	18.63	52.11	14.85	49.63	12.36	30.51	33.48	31.00	50.87	13.61	O.K
114	1.2 DL + 0.8 WL4	72.96	-0.61	11.80	32.43	1.09	20.98	54.50	12.54	52.31	10.36	32.90	33.52	31.34	53.41	11.45	O.K
	Max :	126	21	23													
	Min :	49	-24	-23													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 11.86 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 21.86 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 61.88 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 61.88 kN/m²

Sample calculation of values of base pressure:

Load case : 1 DL + 1 (LL1+LL2+

Maximum pressure, p_1 (pmax) = 46.36+7.22+3.24= 56.82 kN/m²

p_2= 46.36+7.22-3.24= 50.35 kN/m²

p_3= 46.36-(7.22+3.24)= 35.90 kN/m²

p_4= 46.36-7.22+3.24= 42.37 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 56.82-1.2*18= 35.22 kN/m²

Maximum average Pressure along X, $p_{ave(max)}$ = (56.82+50.35)/2= 53.58 kN/m²

Minimum average Pressure along X, $p_{ave(min)}$ = (35.90+42.37)/2= 39.14 kN/m²

Maximum average Pressure along Y, $p_{ave(max)}$ = (56.82+42.37)/2= 49.60 kN/m²

Minimum average Pressure along Y, $p_{ave(min)}$ = (35.90+50.35)/2= 43.13 kN/m²

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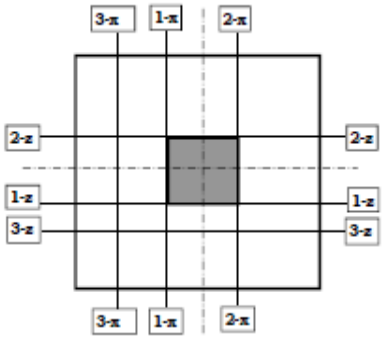
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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= |(0.6+2*0.5*0.426|+|0.6+2*0.5*0.426||*2
= 4.104 m
Punching Area= |(0.6+2*0.5*0.426|*|0.6+2*0.5*0.426|)
= 1.05 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.5*24+0.7*18)
= 24.60 kN/m^2
Area of footing, A_f= 2.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M$_{1x}$ kNm	B.M$_{2x}$ kNm	S.F$_{3x}$ kN	S.F$_{(punching)}$ kN
	Ultimate Conditions							
101	1.4 DL	38.70	32.62	34.51	0.98	0.76	0.08	10.27
102	1.2 (DL) + 1.6 (LL1+LL2+LL3)	60.57	51.68	67.15	4.09	2.28	0.84	37.80
103	1.2 DL+ 1.6 WL1 + 0.5 (LL1+	66.01	32.22	90.91	5.90	-0.94	0.58	29.41
104	1.2 DL+ 1.6 WL2 + 0.5 (LL1+	44.85	19.22	68.78	3.85	-1.84	0.81	6.97
105	1.2 DL+ 1.6 WL3 + 0.5 (LL1+	42.16	39.69	43.99	1.90	1.40	0.16	17.96
106	1.2 DL+ 1.6 WL4 + 0.5 (LL1+	42.04	39.30	44.05	1.90	1.35	0.16	18.42
107	0.9 DL + 1.6 WL1	55.98	24.92	78.87	4.75	-1.54	0.44	19.02
108	0.9 DL + 1.6 WL2	41.17	17.65	58.51	2.87	-1.90	0.27	-8.48
109	0.9 DL + 1.6 WL3	35.02	34.61	35.32	1.08	0.99	0.09	7.57
110	0.9 DL + 1.6 WL4	36.08	35.75	36.25	1.17	1.11	0.09	8.02
111	1.2 DL + 0.8 WL1	45.07	28.96	56.95	2.89	-0.87	0.26	14.89
112	1.2 DL + 0.8 WL2	34.78	20.54	45.28	1.75	-1.13	0.17	3.67
113	1.2 DL + 0.8 WL3	32.74	31.74	33.47	0.87	0.67	0.07	9.16
114	1.2 DL + 0.8 WL4	32.87	31.99	33.51	0.88	0.70	0.07	9.39

Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1x} = 34.52-(34.52-31.8)*0.45/1.5 = 33.70 kN/m^2 Overhang = 0.45 m
p_{2x} = 31.8+(34.52-31.8)*0.45/1.5 = 32.62 kN/m^2 Effective depth = 0.442 m
p_{3x} = 34.52-(34.52-31.8)*(0.45-0.442)/1.5 = 34.51 kN/m^2
BM_{1x} = (33.70-24.6)*0.45*0.45/2+0.5*(34.52-33.70)*0.45*0.45*0.67 = 0.98 kN-m/m
BM_{2x} = (32.62-24.6)*0.45*0.45/2-0.5*(-31.8+32.62)*0.45*0.45*0.67 = 0.76 kN-m/m
SF_{3x} = (34.51-24.6)*(0.45-0.442)+0.5*(34.52-34.51)*(0.45-0.442) = 0.08 kN/m
S.F$_{(punching)}$ = (2.25-1.05)*(33.16-24.6) = 10.2693333 kN

Location	Mux (kN-m)	ϕ Kn = M_{ux}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd. (mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	5.90	0.03	0.04	0.000	42.48	795.60	16φT	252.72	300	200	1005.31
At top	-1.90	-0.01	0.01	0.000	13.61	397.80	16φT	505.43	300	200	1005.31

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4 DL	33.71	32.61	34.49	0.98	0.76	0.24
102	1.2 (DL) + 1.6 (LL1+LL2+LL3)	58.33	46.23	61.49	3.64	1.44	0.89
103	1.2 DL+ 1.6 WL1 + 0.5 (LL1+	50.18	40.22	51.69	2.70	0.40	0.65
104	1.2 DL+ 1.6 WL2 + 0.5 (LL1+	32.20	38.32	32.43	0.79	-0.83	0.19
105	1.2 DL+ 1.6 WL3 + 0.5 (LL1+	56.60	17.81	78.86	4.83	-1.58	1.32
106	1.2 DL+ 1.6 WL4 + 0.5 (LL1+	55.85	16.91	77.41	4.71	1.06	1.28
107	0.9 DL + 1.6 WL1	40.79	39.60	41.27	1.67	1.34	0.40
108	0.9 DL + 1.6 WL2	29.94	38.67	30.68	0.59	0.28	0.15
109	0.9 DL + 1.6 WL3	48.74	17.74	68.52	3.86	-0.11	1.07
110	0.9 DL + 1.6 WL4	50.25	0.00	70.63	4.06	0.00	1.12
111	1.2 DL + 0.8 WL1	37.14	0.00	37.31	1.28	0.00	0.31
112	1.2 DL + 0.8 WL2	28.48	0.00	29.63	0.48	0.00	0.12
113	1.2 DL + 0.8 WL3	39.69	0.00	50.27	2.29	0.00	0.62
114	1.2 DL + 0.8 WL4	40.82	0.00	52.74	2.50	0.00	0.68

Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1z} = 34.58-(34.58-31.79)*0.45/1.5= 33.71 kN/m^2
p_{2z} = 31.79+(34.58-31.79)*0.45/1.5= 32.61 kN/m^2
p_{3z} = 34.58-(34.58-31.79)*(0.45-0.426)/1.5= 34.49 kN/m^2

Overhang = 0.45 m
Effective depth = 0.426 m

BM_{1z} = (33.71-24.6)*0.45*0.45/2+0.5*(34.58-33.71)*0.45*0.45*0.67 = 0.98 kN-m/m
BM_{2z} = (32.61-24.6)*0.45*0.45/2-0.5*|-31.79+32.61|*0.45*0.45*0.67 = 0.76 kN-m/m
SF_{3z} = (34.49-24.6)*(0.45-0.426)+0.5*(34.58-34.49)*(0.45-0.426) = 0.24 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{ux}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd. (mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	4.88	0.027	0.036	0.0001	37.41	766.80	16@T	262.21	300	200	1005.3
At top	-1.58	-0.009	-0.012	0.0000	-0.12	388.40	16@T	524.42	300	200	1005.3

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Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 0.58 kN/m

Shear stress , τ_v= (0.58*1000)/(1000*442)

= 0.00 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 87.80 kN

Strength of Concrete in shear Vc1 = 5365.79 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 8714.91 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 8577.19 kN **Table 22.6.5.2_ (c)**

Design Vc= 8577.19 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (87.8*1000)/(4104*442)

= 0.02 N/mm^2 **No shear reinforcement reqd.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	16ϕT	200	c/c
Along Z Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	16ϕT	200	c/c

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5.3 Design of Foundation F7 & F8:

	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	3.00m	3.00m
Thickness of raft :	0.50 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.55 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

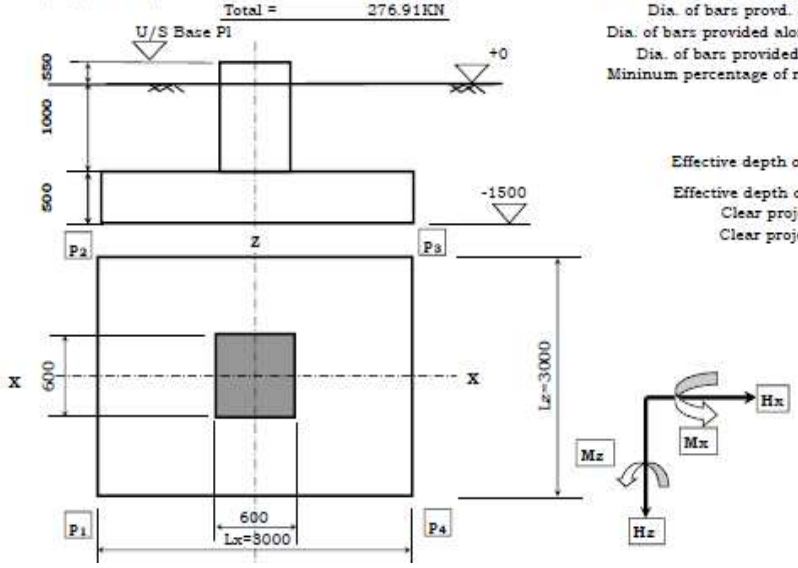
Calculation of Weight of Foundation :

Weight of raft & Col.= (3*3*0.5*24)+(0.6*0.6*1.55*24) = 121.39KN
Weight of soil = ((3*3)-(0.6*0.6))*1*18 = 155.52KN
Total = 276.91KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 442mm = 0.442m
Effective depth of slab along Z direction = 426mm = 0.426m
Clear projection along Z direction = 1.2m
Clear projection along X direction = 1.2m



H_X=Force along X-X direction
Hz=Force along Z-Z direction
M_X=Moment about X-X direction
Mz=Moment about Z-Z direction

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Combined Load Cases:	Load at top of foundation					Load at bottom of foundation				
Load Case / Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 1. DL	103.1	-3.2	0.0	28.0	0.0	380.05	-3.17	-6.49	28.00	57.40
202 1 DL + 1 (LL1+LL2+LL3)	490.6	-8.4	0.0	145.1	0.0	767.55	-8.37	-17.17	145.15	297.55
203 1 DL+ 1 WL1	151.4	-40.0	0.0	27.5	0.0	428.33	-40.01	-82.02	27.49	56.35
204 1 DL+ 1 WL2	54.9	33.7	0.0	28.5	0.0	331.76	33.68	69.04	28.51	58.44
205 1 DL+ 1 WL3	32.1	-6.0	0.0	-32.7	0.0	309.02	-5.97	-12.23	-32.68	-66.99
206 1 DL+ 1 WL4	174.2	-0.4	0.0	88.7	0.0	451.07	-0.36	-0.74	88.68	181.79
207 0.6 DL + 1 WL1	110.2	-38.7	0.0	16.3	0.0	387.08	-38.74	-79.43	16.29	33.39
208 0.6 DL + 1 WL2	13.6	34.9	0.0	17.3	0.0	290.51	34.95	71.64	17.31	35.48
209 0.6 DL + 1 WL3	-9.1	-4.7	0.0	-43.9	0.0	267.77	-4.70	-9.64	-43.88	-89.95
210 0.6 DL + 1 WL4	132.9	0.9	0.0	77.5	0.0	409.82	0.90	1.85	77.48	158.83
211 1 DL+ 0.75 WL1+ 0.75 (LL1+LL2+LL3)	430.0	-34.7	0.0	115.5	0.0	706.89	-34.71	-71.15	115.48	236.73
212 1 DL+ 0.75 WL2+ 0.75 (LL1+LL2+LL3)	357.6	20.6	0.0	116.2	0.0	634.46	20.56	42.15	116.24	238.30
213 1 DL+ 0.75 WL3+ 0.75 (LL1+LL2+LL3)	340.5	-9.2	0.0	70.4	0.0	617.41	-9.17	-18.80	70.35	144.22
214 1 DL+ 0.75 WL4+ 0.75 (LL1+LL2+LL3)	447.0	-5.0	0.0	161.4	0.0	723.94	-4.97	-10.19	161.37	330.80
Ultimate Conditions										
101 1.4 DL	144.4	-4.4	0.0	39.2	0.0	421.30	-4.43	-9.08	39.20	80.36
102 1.2 (DL) + 1.6 (LL1+LL2+LL3)	743.8	-12.1	0.0	221.0	0.0	1020.68	-12.13	-24.87	221.04	453.12
103 1.2 DL+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)	394.8	-65.4	0.0	91.4	0.0	671.68	-65.36	-133.98	91.36	187.28
104 1.2 DL+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)	240.3	52.5	0.0	93.0	0.0	517.17	52.55	107.73	92.99	190.63
105 1.2 DL+ 1.6 WL3 + 0.5 (LL1+LL2+LL3)	203.9	-10.9	0.0	-4.9	0.0	480.79	-10.89	-22.32	-4.91	-10.07
106 1.2 DL+ 1.6 WL4 + 0.5 (LL1+LL2+LL3)	431.2	-1.9	0.0	189.3	0.0	708.06	-1.92	-3.94	189.26	387.98
107 0.9 DL + 1.6 WL1	170.1	-61.8	0.0	24.4	0.0	446.99	-61.80	-126.69	24.38	49.99
108 0.9 DL + 1.6 WL2	15.6	56.1	0.0	26.0	0.0	292.48	56.10	115.01	26.02	53.33
109 0.9 DL + 1.6 WL3	-20.8	-7.3	0.0	-71.9	0.0	256.10	-7.33	-15.03	-71.88	-147.36
110 0.9 DL + 1.6 WL4	206.5	1.6	0.0	122.3	0.0	483.37	1.63	3.35	122.28	250.68
111 1.2 DL + 0.8 WL1	162.4	-33.3	0.0	33.2	0.0	439.30	-33.27	-68.21	33.19	68.04
112 1.2 DL + 0.8 WL2	85.1	25.7	0.0	34.0	0.0	362.05	25.68	52.64	34.01	69.71
113 1.2 DL + 0.8 WL3	66.9	-6.0	0.0	-14.9	0.0	343.86	-6.04	-12.38	-14.94	-30.63
114 1.2 DL + 0.8 WL4	180.6	-1.6	0.0	82.1	0.0	457.50	-1.56	-3.19	82.14	168.39

Check for overturning and sliding:

Load Case / Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	380.0	-3.2	-6.5	28.0	57.4	87.9	9.9	48.0	5.4
202 1 DL + 1 (LL1+LL2+LL3)	767.5	-8.4	-17.2	145.1	297.6	67.1	3.9	36.7	2.1
203 1 DL+ 1 WL1	428.3	-40.0	-82.0	27.5	56.4	7.8	11.4	4.3	6.2
204 1 DL+ 1 WL2	331.8	33.7	69.0	28.5	58.4	7.2	8.5	3.9	4.7
205 1 DL+ 1 WL3	309.0	-6.0	-12.2	-32.7	-67.0	37.9	6.9	20.7	3.8
206 1 DL+ 1 WL4	451.1	-0.4	-0.7	88.7	181.8	909.2	3.7	497.0	2.0
207 0.6 DL + 1 WL1	387.1	-38.7	-79.4	16.3	33.4	7.3	17.4	4.0	9.5
208 0.6 DL + 1 WL2	290.5	34.9	71.6	17.3	35.5	6.1	12.3	3.3	6.7
209 0.6 DL + 1 WL3	267.8	-4.7	-9.6	-43.9	-89.9	41.7	4.5	22.8	2.4
210 0.6 DL + 1 WL4	409.8	0.9	1.9	77.5	158.8	332.1	3.9	181.5	2.1
211 1 DL+ 0.75 WL1+ 0.75 (LL1+LL2+LL3)	706.9	-34.7	-71.1	115.5	236.7	14.9	4.5	8.1	2.4
212 1 DL+ 0.75 WL2+ 0.75 (LL1+LL2+LL3)	634.5	20.6	42.2	116.2	238.3	22.6	4.0	12.3	2.2
213 1 DL+ 0.75 WL3+ 0.75 (LL1+LL2+LL3)	617.4	-9.2	-18.8	70.4	144.2	49.2	6.4	26.9	3.5
214 1 DL+ 0.75 WL4+ 0.75 (LL1+LL2+LL3)	723.9	-5.0	-10.2	161.4	330.8	106.6	3.3	58.3	1.8
					Min.=	6.1	3.3	3.3	1.8
						OK	OK	OK	OK

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DESIGN OF FOUNDATION MKD. F7

Load Combination & Calculation of Base pressure

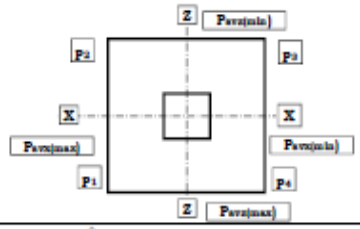
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mx/((Lx)^2.Lz)) + (6.Mx/(Lx.(Lz)^2))$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mx/((Lx)^2.Lz)) - (6.Mx/(Lx.(Lz)^2))$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 450 kN/m² For WL/SL case = 562.5 kN/m²

Gross SBC of soil for DL, LL= 477 kN/m² For WL/SL case = 589.5 kN/m²



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mx (kN-m)	Mz (kN-m)	V/(Lx.Lz)	6.Mx/(Lx)².Lz	6.Mz/Lx.(Lz)²	Base Pressure (kN/m²) P1 (Pmax)	P2	P4	P3 (Pmin)	(Pmax-Pb)	Pave(max)	Pave(min)	Pave(max)	Pave(min)	Remarks
201	1. DL	380.05	-6.49	57.40	42.23	1.44	12.76	56.42	30.91	53.54	28.03	29.42	43.67	40.79	54.98	29.47	O.K
202	1 DL + 1 (LL1+LL2+LL3)	767.55	-17.17	297.55	85.28	3.81	66.12	155.22	22.98	147.59	15.35	126.22	89.10	81.47	151.41	19.17	O.K
203	1 DL+ 1 WL1	428.33	-82.02	56.35	47.59	18.23	12.52	78.34	53.30	41.89	16.84	51.34	65.82	29.37	60.12	35.07	O.K
204	1 DL+ 1 WL2	331.76	69.04	58.44	36.86	15.34	12.99	65.19	39.22	34.51	8.53	38.19	52.21	21.52	49.85	23.88	O.K
205	1 DL+ 1 WL3	309.02	-12.23	-66.99	34.34	2.72	14.89	51.94	22.17	46.50	16.73	24.94	37.06	31.62	49.22	19.45	O.K
206	1 DL+ 1 WL4	451.07	-0.74	181.79	50.12	0.17	40.40	90.68	9.89	90.35	9.56	63.68	50.29	49.96	90.52	9.73	O.K
207	0.6 DL + 1 WL1	387.08	-79.43	33.39	43.01	17.65	7.42	68.08	53.24	32.78	17.94	41.08	60.66	25.36	50.43	35.59	O.K
208	0.6 DL + 1 WL2	290.51	71.64	35.48	32.28	15.92	7.89	56.08	40.31	24.24	8.47	29.08	48.20	16.36	40.16	24.39	O.K
209	0.6 DL + 1 WL3	267.77	-9.64	-89.95	29.75	2.14	19.99	51.88	11.91	47.60	7.62	24.88	31.90	27.61	49.74	9.77	O.K
210	0.6 DL + 1 WL4	409.82	1.85	158.83	45.54	0.41	35.30	81.24	10.65	80.42	9.83	54.24	45.95	45.13	80.83	10.24	O.K
211	1 DL+ 0.75 WL1+ 0.75 (LL1+LL2	706.89	-71.15	236.73	78.54	15.81	52.61	146.96	41.75	115.34	10.13	119.96	94.36	62.74	131.15	25.94	O.K
212	1 DL+ 0.75 WL2+ 0.75 (LL1+LL2	634.46	42.15	238.30	70.50	9.37	52.95	132.82	26.91	114.08	8.17	105.82	79.87	61.13	123.45	17.54	O.K
213	1 DL+ 0.75 WL3+ 0.75 (LL1+LL2	617.41	-18.80	144.22	68.60	4.18	32.05	104.83	40.73	96.47	32.37	77.83	72.78	64.42	100.65	36.55	O.K
214	1 DL+ 0.75 WL4+ 0.75 (LL1+LL2	723.94	-10.19	330.80	80.44	2.26	73.51	156.21	9.19	151.69	4.66	129.21	82.70	78.18	153.95	6.93	O.K
								Ultimate Conditions									
101	1.4 DL	421.30	-9.08	80.36	46.81	2.02	17.86	66.69	30.97	62.65	26.94	39.69	48.83	44.80	64.67	28.96	
102	1.2 (DL) + 1.6 (LL1+LL2+LL3)	1020.68	-24.87	453.12	113.41	5.53	100.69	219.63	18.24	208.58	7.19	192.63	118.94	107.89	214.11	12.72	
103	1.2 DL+ 1.6 WL1 + 0.5 (LL1+LL2	671.68	-133.98	187.28	74.63	29.77	41.62	146.02	62.79	86.48	3.24	119.02	104.41	44.86	116.25	33.02	
104	1.2 DL+ 1.6 WL2 + 0.5 (LL1+LL2	517.17	107.73	190.63	57.46	23.94	42.36	123.76	39.04	75.89	-8.84	96.76	81.40	33.53	99.83	15.10	
105	1.2 DL+ 1.6 WL3 + 0.5 (LL1+LL2	480.79	-22.32	-10.07	53.42	4.96	2.24	60.62	56.14	50.70	46.22	33.62	58.38	48.46	55.66	51.18	
106	1.2 DL+ 1.6 WL4 + 0.5 (LL1+LL2	708.06	-3.94	387.98	78.67	0.87	86.22	165.77	-6.67	164.02	-8.42	138.77	79.55	77.80	164.90	-7.55	
107	0.9 DL + 1.6 WL1	446.99	-126.69	49.99	49.67	28.15	11.11	88.93	66.71	32.62	10.40	61.93	77.82	21.51	60.78	38.56	
108	0.9 DL + 1.6 WL2	292.48	115.01	53.33	32.50	25.56	11.85	69.91	46.20	18.79	-4.91	42.91	58.06	6.94	44.35	20.65	
109	0.9 DL + 1.6 WL3	256.10	-15.03	-147.36	28.46	3.34	32.75	64.54	-0.95	57.86	-7.63	37.54	31.80	25.12	61.20	-4.29	
110	0.9 DL + 1.6 WL4	483.37	3.35	250.68	53.71	0.74	55.71	110.16	-1.25	108.67	-2.74	83.16	54.46	52.97	109.42	-2.00	
111	1.2 DL + 0.8 WL1	439.30	-68.21	68.04	48.81	15.16	15.12	79.09	48.85	48.77	18.53	52.09	63.97	33.65	63.93	33.69	
112	1.2 DL + 0.8 WL2	362.05	52.64	69.71	40.23	11.70	15.49	67.42	36.43	44.02	13.04	40.42	51.93	28.53	55.72	24.74	
113	1.2 DL + 0.8 WL3	343.86	-12.38	-30.63	38.21	2.75	6.81	47.76	34.15	42.26	28.65	20.76	40.96	35.46	45.01	31.40	
114	1.2 DL + 0.8 WL4	457.50	-3.19	168.39	50.83	0.71	37.42	88.96	14.12	87.54	12.70	61.96	51.54	50.12	88.25	13.41	
	Max :	1021	115	453													
	Min :	256	-134	-147													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 29.42 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 39.42 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 129.21 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 139.21 kN/m²

Sample calculation of values of base pressure:

Load case : 1 DL + 1 (LL1+LL2+

Maximum pressure, p_1 (pmax) = 85.28+3.81+66.12= 155.22 kN/m²

p_2= 85.28+3.81-66.12= 22.98 kN/m²

p_3= 85.28-(3.81+66.12)= 15.35 kN/m²

p_4= 85.28-3.81+66.12= 147.59 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γb)= 155.22-1.5*18= 128.22 kN/m²

Maximum average Pressure along X, $p_{ave(max)}$ = (155.22+22.98)/2= 89.10 kN/m²

Minimum average Pressure along X, $p_{ave(min)}$ = (15.35+147.59)/2= 81.47 kN/m²

Maximum average Pressure along Y, $p_{ave(max)}$ = (155.22+147.59)/2= 151.41 kN/m²

Minimum average Pressure along Y, $p_{ave(min)}$ = (15.35+22.98)/2= 19.16 kN/m²

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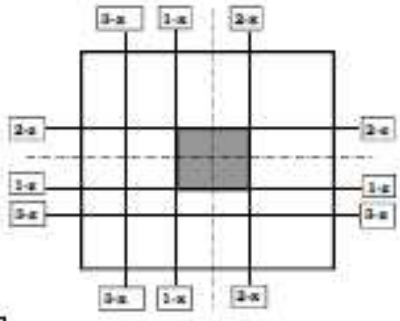
Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m².
p_{2x} =pressure at the face of column at section 2x-2x in kN/m².
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m².
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= (0.6+2*0.5*0.426)+(0.6+2*0.5*0.426)*2
= 6.104 m

Punching Area= (0.6+2*0.5*0.426)*(0.6+2*0.5*0.426))
= 1.05 m² Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.5*24+1*18)
= 30.00 kN/m²
Area of footing, A_f= 9.00 m²



Load Case	Load Combination	P_{1x} kN/m2	P_{2x} kN/m2	P_{3x} kN/m2	BM_{1x} kNm	BM_{2x} kNm	SF_{3x} kN	$S.F_{(punching)}$ kN
Ultimate Conditions								
101	1.4 DL	47.22	46.41	47.81	13.17	11.04	13.89	133.65
102	1.2 (DL) + 1.6 (LL1+LL2+LL3)	114.52	112.31	116.15	62.98	57.14	66.36	663.10
103	1.2 DL+ 1.6 WL1 + 0.5 (LL1+L	80.59	68.68	89.86	47.86	35.42	50.70	354.82
104	1.2 DL+ 1.6 WL2 + 0.5 (LL1+L	62.23	52.68	69.30	32.41	7.14	34.38	218.33
105	1.2 DL+ 1.6 WL3 + 0.5 (LL1+L	54.41	52.43	55.87	19.48	14.24	20.26	186.20
106	1.2 DL+ 1.6 WL4 + 0.5 (LL1+L	78.80	78.50	79.11	35.51	34.58	37.29	386.96
107	0.9 DL + 1.6 WL1	55.30	44.03	63.59	29.02	-0.71	30.86	156.34
108	0.9 DL + 1.6 WL2	37.61	27.39	45.14	15.30	-11.70	16.37	19.85
109	0.9 DL + 1.6 WL3	29.13	27.79	30.11	0.65	-2.87	0.72	-12.28
110	0.9 DL + 1.6 WL4	53.86	53.57	54.08	17.47	15.68	18.40	168.48
111	1.2 DL + 0.8 WL1	51.84	45.78	56.31	21.55	5.54	22.85	149.55
112	1.2 DL + 0.8 WL2	42.57	37.89	46.02	13.54	1.19	14.38	81.31
113	1.2 DL + 0.8 WL3	38.76	37.66	39.57	7.36	4.46	7.78	65.24
114	1.2 DL + 0.8 WL4	50.97	50.69	51.18	15.37	14.62	16.19	165.62

Load case : 101 1.4 DL

P_{1x} = 48.83-(48.83-44.8)*1.2/3 = 47.22 kN/m²
P_{2x} = 44.8+(48.83-44.8)*1.2/3 = 46.41 kN/m²
P_{3x} = 48.83-(48.83-44.8)*(1.2-0.442)/3 = 47.81 kN/m²
BM_{1x} = (47.22-30)*1.2*1.2/2+0.5*(48.83-47.22)*1.2*1.2*0.67 = 13.17 kN-m/m
BM_{2x} = (46.41-30)*1.2*1.2/2-0.5*(-44.8+46.41)*1.2*1.2*0.67 = 11.04 kN-m/m
SF_{3x} = (47.81-30)*(1.2-0.442)+0.5*(48.83-47.81)*(1.2-0.442) = 13.89 kN/m
$S.F_{(punching)}$ = .05)*(46.81-30) = 133.650983 kN

Overhang = 1.2 m
Effective depth = 0.442 m

Location	Mux (kN-m)	ϕ Kn = M_{ux}/bd^2 (N/mm²)	Kn (N/mm²)	ρ=	Ast (mm²)	Minm. Ast (mm²)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm²)
At bottom	62.98	0.32	0.43	0.001	455.63	795.60	16@T	252.72	300	200	1005.31
At top	-11.70	-0.06	0.08	0.000	84.12	397.80	10@T	197.44	300	200	392.70

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	P_{1z} kN/m2	P_{2z} kN/m2	P_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4 DL	50.39	43.24	55.46	21.57	2.65	23.27
102	1.2 (DL) + 1.6 (LL1+LL2+LL3)	133.55	27.80	162.15	113.42	2.17	122.39
103	1.2 DL+ 1.6 WL1 + 0.5 (LL1+L	82.96	35.88	94.78	54.19	-1.31	58.45
104	1.2 DL+ 1.6 WL2 + 0.5 (LL1+L	65.94	28.96	77.97	42.22	-10.01	45.59
105	1.2 DL+ 1.6 WL3 + 0.5 (LL1+L	53.87	63.04	54.50	18.05	-1.68	19.41
106	1.2 DL+ 1.6 WL4 + 0.5 (LL1+L	95.92	47.93	120.41	80.74	2.30	87.19
107	0.9 DL + 1.6 WL1	51.89	72.52	55.05	20.05	4.09	21.61
108	0.9 DL + 1.6 WL2	34.87	52.65	38.24	8.08	8.54	8.74
109	0.9 DL + 1.6 WL3	35.00	59.01	44.30	16.24	-4.24	17.61
110	0.9 DL + 1.6 WL4	64.85		80.67	46.59	0.00	50.35
111	1.2 DL + 0.8 WL1	51.83	0.00	56.13	21.56	0.00	23.24
112	1.2 DL + 0.8 WL2	43.33	0.00	47.73	15.57	0.00	16.81
113	1.2 DL + 0.8 WL3	39.57	0.00	41.50	9.51	0.00	10.26

Sample calculation of BM & SF

Load case : 101 1.4 DL

P_{1z} = 64.67-(64.67-28.96)*1.2/3= 50.39 kN/m^2
P_{2z} = 28.96+(64.67-28.96)*1.2/3= 43.24 kN/m^2
P_{3z} = 64.67-(64.67-28.96)*(1.2-0.426)/3= 55.46 kN/m^2
BM_{1z} = (50.39-30)*1.2*1.2/2+0.5*(64.67-50.39)*1.2*1.2*0.67 = 21.57 kN-m/m
BM_{2z} = (43.24-30)*1.2*1.2/2-0.5*(-28.96+43.24)*1.2*1.2*0.67 = 2.65 kN-m/m
SF_{3z} = (55.46-30)*(1.2-0.426)+0.5*(64.67-55.46)*(1.2-0.426) = 23.27 kN/m

Overhang = 1.2 m
Effective depth = 0.426 m

Location	Mux (kN-m)	ϕ Kn = M_{ux}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	113.42	0.625	0.833	0.0020	889.60	766.80	16@T	226.01	300	200	1005.3
At top	-10.01	-0.055	-0.074	-0.0002	-0.74	383.40	10@T	204.85	300	200	392.7

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Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi Vc)/bd$ = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 66.36 kN/m

Shear stress , τ_v= (66.36*1000)/(1000*442)

= 0.15 N/mm^2 O.K.

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 663.10 kN

Strength of Concrete in shear Vc1 = 5365.79 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 3714.91 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 3577.19 kN Table 22.6.5.2_ (c)

Design Vc= 3577.19 kN

Allowable shear stress, $(\phi Vc)/bd$ = 1.48 N/mm^2

Design punching shear stress, τ = (663.1*1000)/(4104*442)

= 0.37 N/mm^2 No shear reinforcement reqd.

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	10ϕT	200	c/c
Along Z Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	10ϕT	200	c/c

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5.4 Design of Foundation F9:

Footing F9 is in close proximity of gantry foundation. Hence, bottom of this foundation placed below gantry foundation and an eccentric surcharge load of 3T/sqm is considered for design purpose.

Forces at bottom of the foundation is calculated from STAAD (see attachment) and design is performed in excel.

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DESIGN OF FOOTING TYPE F9

	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	3.00m	3.00m
Thickness of raft :	0.60	m
Original ground level:	0.00	m
Finished floor level :	0.00	m
U/S level of foundation :	-2.40	m
U/S level Base Plate :	0.55	m
Unit weight of conc :	24	kN/m^3
Unit weight of soil :	18	kN/m^3

SYMBOLS AND NOTATIONS

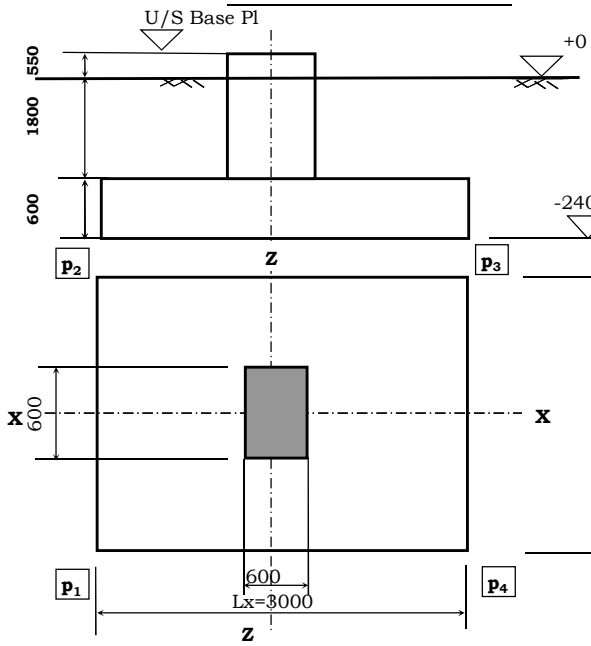
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

ght of raft & Col.=	(3*3*0.6*24)+(0.6*0.6*2.35*24) =	149.90KN
Weight of soil =	((3*3)-(0.6*0.6))*1.8*18 =	279.94KN
	Total =	429.84KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	20mm
Dia. of bars provd. along X direction at top =	20mm
Dia. of bars provided along Z direction at bottom=	20mm
Dia. of bars provided along Z direction at top=	20mm
Mininum percentage of reinforcement at bottom=	0.18%

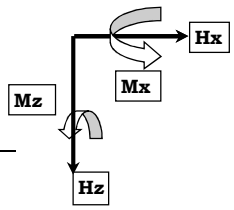


Effective depth of slab along X direction = 540mm = 0.54m

Effective depth of slab along Z direction = 520mm = 0.52m

Clear projection along Z direction = 1.2m

Clear projection along X direction = 1.2m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Sensitivity: LNT Construction Internal Use

DESIGN OF FOOTING TYPE F9

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
201 1. DL	338.6	12.3	-133.3	5.1	13.3	768.46	12.33	-133.30	5.10	13.26
202 1 (DL+TL1) + 1 LL	716.8	36.5	-196.2	7.4	19.3	1146.68	36.53	-196.22	7.42	19.29
203 1 (DL+TL2) + 1 LL	716.8	36.5	-196.2	7.4	19.3	1146.68	36.53	-196.22	7.42	19.29
204 1 DL+ 1 WL(+X)	328.3	2.1	-106.7	4.6	11.9	758.10	2.09	-106.69	4.58	11.90
205 1 DL+ 1 WL(-X)	349.0	22.6	-159.9	4.6	11.9	778.81	22.56	-159.90	4.58	11.90
206 1 DL+ 1 WL(+Z)	235.3	8.1	-122.3	-14.7	-38.2	665.17	8.08	-122.26	-14.71	-38.25
207 1 DL+ 1 WL(-Z)	441.9	16.6	-144.3	24.9	64.8	871.75	16.57	-144.33	24.91	64.77
208 0.6 DL + 1 WL(+X)	192.8	-2.8	-53.4	2.5	6.6	622.65	-2.84	-53.37	2.54	6.59
209 0.6 DL + 1 WL(-X)	213.5	17.6	-106.6	2.5	6.6	643.37	17.63	-106.58	2.54	6.59
210 0.6 DL + 1 WL(+Z)	99.9	3.2	-68.9	-16.8	-43.6	529.72	3.15	-68.95	-16.75	-43.55
211 0.6 DL + 1 WL(-Z)	306.5	11.6	-91.0	22.9	59.5	736.30	11.64	-91.01	22.87	59.47
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	614.5	22.8	-160.5	6.4	16.8	1044.36	22.80	-160.53	6.45	16.76
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	630.1	38.1	-200.4	6.4	16.8	1059.90	38.15	-200.44	6.45	16.76
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	544.8	27.3	-172.2	-8.0	-20.8	974.66	27.29	-172.21	-8.02	-20.85
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	699.8	33.7	-188.8	21.7	56.4	1129.60	33.66	-188.76	21.70	56.42
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00

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Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
101 1.4 DL	474.1	17.3	-186.6	7.1	18.6	903.90	17.26	-186.61	7.14	18.56
102 1.2 (DL+TL1) + 1.6 LL	1011.5	53.5	-260.6	9.8	25.6	1441.34	53.51	-260.63	9.83	25.56
103 1.2 (DL+TL2) + 1.6 LL	1011.5	53.5	-260.6	9.8	25.6	1441.34	53.51	-260.63	9.83	25.56
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	578.9	10.5	-148.8	6.4	16.7	1008.72	10.52	-148.85	6.44	16.74
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	612.0	43.3	-234.0	6.4	16.7	1041.87	43.26	-233.98	6.44	16.74
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	430.2	20.1	-173.8	-24.4	-63.5	860.03	20.10	-173.76	-24.42	-63.49
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	760.7	33.7	-209.1	39.0	101.3	1190.56	33.68	-209.07	38.98	101.34
108 0.9 DL + 1.6 WL(+X)	288.2	-5.3	-77.4	3.8	9.8	718.02	-5.28	-77.40	3.75	9.75
109 0.9 DL + 1.6 WL(-X)	321.3	27.5	-162.5	3.8	9.8	751.17	27.46	-162.53	3.75	9.75
110 0.9 DL + 1.6 WL(+Z)	139.5	4.3	-102.3	-27.1	-70.5	569.33	4.30	-102.32	-27.11	-70.48
111 0.9 DL + 1.6 WL(-Z)	470.0	17.9	-137.6	36.3	94.3	899.86	17.88	-137.62	36.29	94.35
112 1.2 DL + 0.8 WL(+X)	398.1	6.6	-138.7	5.7	14.8	827.90	6.60	-138.67	5.70	14.82
113 1.2 DL + 0.8 WL(-X)	414.6	23.0	-181.2	5.7	14.8	844.47	22.98	-181.24	5.70	14.82
114 1.2 DL + 0.8 WL(+Z)	323.7	11.4	-151.1	-9.7	-25.3	753.55	11.40	-151.13	-9.73	-25.30
115 1.2 DL + 0.8 WL(-Z)	323.7	11.4	-151.1	-9.7	-25.3	753.55	11.40	-151.13	-9.73	-25.30
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00
0	0.0	0.0	0.0	0.0	0.0	429.84	0.00	0.00	0.00	0.00

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Check for overturning and sliding:

						Overturning		Sliding	
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	768.5	12.3	-133.3	5.1	13.3	8.6	86.9	24.9	60.3
202 1 (DL+TL1) + 1 LL	1146.7	36.5	-196.2	7.4	19.3	8.8	89.2	12.6	61.8
203 1 (DL+TL2) + 1 LL	1146.7	36.5	-196.2	7.4	19.3	8.8	89.2	12.6	61.8
204 1 DL+ 1 WL(+X)	758.1	2.1	-106.7	4.6	11.9	10.7	95.6	144.9	66.3
205 1 DL+ 1 WL(-X)	778.8	22.6	-159.9	4.6	11.9	7.3	98.2	13.8	68.1
206 1 DL+ 1 WL(+Z)	665.2	8.1	-122.3	-14.7	-38.2	8.2	26.1	32.9	18.1
207 1 DL+ 1 WL(-Z)	871.7	16.6	-144.3	24.9	64.8	9.1	20.2	21.0	14.0
208 0.6 DL + 1 WL(+X)	622.7	-2.8	-53.4	2.5	6.6	17.5	141.7	87.8	98.2
209 0.6 DL + 1 WL(-X)	643.4	17.6	-106.6	2.5	6.6	9.1	146.4	14.6	101.5
210 0.6 DL + 1 WL(+Z)	529.7	3.2	-68.9	-16.8	-43.6	11.5	18.2	67.2	12.6
211 0.6 DL + 1 WL(-Z)	736.3	11.6	-91.0	22.9	59.5	12.1	18.6	25.3	12.9
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	1044.4	22.8	-160.5	6.4	16.8	9.8	93.5	18.3	64.8
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	1059.9	38.1	-200.4	6.4	16.8	7.9	94.9	11.1	65.8
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	974.7	27.3	-172.2	-8.0	-20.8	8.5	70.1	14.3	48.6
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	1129.6	33.7	-188.8	21.7	56.4	9.0	30.0	13.4	20.8
0	429.8	0.0	0.0	0.0	0.0	NA	NA	NA	NA
0	429.8	0.0	0.0	0.0	0.0	NA	NA	NA	NA
0	429.8	0.0	0.0	0.0	0.0	NA	NA	NA	NA
0	429.8	0.0	0.0	0.0	0.0	NA	NA	NA	NA
0	429.8	0.0	0.0	0.0	0.0	NA	NA	NA	NA
					Min.=	7.3	18.2	11.1	12.6
						OK	OK	OK	OK

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DESIGN OF FOOTING TYPE F9

Load Combination & Calculation of Base pressure

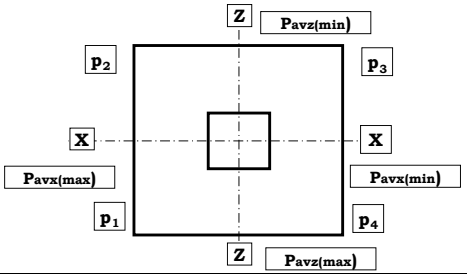
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 450 kN/m^2 For WL/SL case = 562.5 kN/m^2

Gross SBC of soil for DL, LL= 493.2 kN/m^2 For WL/SL case = 605.7 kN/m^2



Allowable Stress Design																	
Load Combination	Design Load/moments			Base Pressure (kN/m^2)													
	V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201 1. DL	768.46	-133.30	13.26	85.38	29.62	2.95	117.95	112.06	58.71	52.82	74.75	115.01	55.77	88.33	82.44	O.K
202 1 (DL+TL1) + 1 LL	1146.68	-196.22	19.29	127.41	43.60	4.29	448.00	166.73	88.09	79.52	404.80	307.37	83.81	268.05	123.13	O.K
203 1 (DL+TL2) + 1 LL	1146.68	-196.22	19.29	127.41	43.60	4.29	448.00	166.73	88.09	79.52	404.80	307.37	83.81	268.05	123.13	O.K
204 1 DL+ 1 WL(+X)	758.10	-106.69	11.90	84.23	23.71	2.64	110.59	105.30	63.17	57.88	67.39	107.95	60.53	86.88	81.59	O.K
205 1 DL+ 1 WL(-X)	778.81	-159.90	11.90	86.53	35.53	2.64	124.71	119.42	53.65	48.36	81.51	122.07	51.01	89.18	83.89	O.K
206 1 DL+ 1 WL(+Z)	665.17	-122.26	-38.25	73.91	27.17	8.50	260.00	92.58	55.24	38.24	216.80	176.29	46.74	157.62	65.41	O.K
207 1 DL+ 1 WL(-Z)	871.75	-144.33	64.77	96.86	32.07	14.39	143.33	114.54	79.18	50.40	100.13	128.94	64.79	111.26	82.47	O.K
208 0.6 DL + 1 WL(+X)	622.65	-53.37	6.59	69.18	11.86	1.46	82.51	79.58	58.79	55.86	39.31	81.05	57.33	70.65	67.72	O.K
209 0.6 DL + 1 WL(-X)	643.37	-106.58	6.59	71.49	23.68	1.46	96.63	93.71	49.27	46.34	53.43	95.17	47.81	72.95	70.03	O.K
210 0.6 DL + 1 WL(+Z)	529.72	-68.95	-43.55	58.86	15.32	9.68	200.00	64.50	53.22	33.86	156.80	132.25	43.54	126.61	49.18	O.K
211 0.6 DL + 1 WL(-Z)	736.30	-91.01	59.47	81.81	20.22	13.21	115.25	88.82	74.80	48.37	72.05	102.04	61.59	95.03	68.60	O.K
212 1 DL+ 0.75 WL(+X)+ 0.75 LL	1044.36	-160.53	16.76	116.04	35.67	3.72	300.00	147.99	84.09	76.64	256.80	224.00	80.37	192.05	112.32	O.K
213 1 DL+ 0.75 WL(-X)+ 0.75 LL	1059.90	-200.44	16.76	117.77	44.54	3.72	407.00	158.58	76.95	69.50	363.80	282.79	73.23	241.98	114.04	O.K
214 1 DL+ 0.75 WL(+Z)+ 0.75 LL	974.66	-172.21	-20.85	108.30	38.27	4.63	460.00	141.93	0.00	65.39	416.80	300.97	32.70	230.00	103.66	O.K
215 1 DL+ 0.75 WL(-Z)+ 0.75 LL	1129.60	-188.76	56.42	125.51	41.95	125.00	384.00	42.46	208.56	0.00	340.80	213.23	104.28	296.28	21.23	O.K
0	429.84	0.00	0.00	47.76	0.00	0.00	47.76	47.76	47.76	47.76	4.56	47.76	47.76	47.76	47.76	O.K
0	429.84	0.00	0.00	47.76	0.00	0.00	47.76	47.76	47.76	47.76	4.56	47.76	47.76	47.76	47.76	O.K
0	429.84	0.00	0.00	47.76	0.00	0.00	47.76	47.76	47.76	47.76	4.56	47.76	47.76	47.76	47.76	O.K
0	429.84	0.00	0.00	47.76	0.00	0.00	47.76	47.76	47.76	47.76	4.56	47.76	47.76	47.76	47.76	O.K
0	429.84											47.76	47.76	47.76	47.76	O.K

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Strength Design

101 1.4 DL	903.90	-186.61	18.56	100.43	41.47	4.13	146.03	137.78	63.09	54.84	102.83	141.91	58.97	104.56	96.31	
102 1.2 (DL+TL1) + 1.6 LL	1441.34	-260.63	25.56	160.15	57.92	5.68	223.75	212.39	107.91	96.55	180.55	218.07	102.23	165.83	154.47	
103 1.2 (DL+TL2) + 1.6 LL	1441.34	-260.63	25.56	160.15	57.92	5.68	223.75	212.39	107.91	96.55	180.55	218.07	102.23	165.83	154.47	
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	1008.72	-148.85	16.74	112.08	33.08	3.72	148.88	141.44	82.72	75.28	105.68	145.16	79.00	115.80	108.36	
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	1041.87	-233.98	16.74	115.76	52.00	3.72	171.48	164.04	67.49	60.05	128.28	167.76	63.77	119.49	112.05	
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	860.03	-173.76	-63.49	95.56	38.61	14.11	148.28	120.06	71.05	42.84	105.08	134.17	56.95	109.67	81.45	
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	1190.56	-209.07	101.34	132.28	46.46	22.52	201.26	156.22	108.35	63.30	158.06	178.74	85.83	154.81	109.76	
108 0.9 DL + 1.6 WL(+X)	718.02	-77.40	9.75	79.78	17.20	2.17	99.15	94.81	64.75	60.41	55.95	96.98	62.58	81.95	77.61	
109 0.9 DL + 1.6 WL(-X)	751.17	-162.53	9.75	83.46	36.12	2.17	121.75	117.41	49.51	45.18	78.55	119.58	47.35	85.63	81.30	
110 0.9 DL + 1.6 WL(+Z)	569.33	-102.32	-70.48	63.26	22.74	15.66	101.66	70.33	56.18	24.86	58.46	86.00	40.52	78.92	47.60	
111 0.9 DL + 1.6 WL(-Z)	899.86	-137.62	94.35	99.98	30.58	20.97	151.53	109.60	90.37	48.44	108.33	130.57	69.41	120.95	79.02	
112 1.2 DL + 0.8 WL(+X)	827.90	-138.67	14.82	91.99	30.82	3.29	126.10	119.51	64.47	57.88	82.90	122.81	61.18	95.29	88.70	
113 1.2 DL + 0.8 WL(-X)	844.47	-181.24	14.82	93.83	40.27	3.29	137.40	130.81	56.85	50.26	94.20	134.11	53.56	97.13	90.54	
114 1.2 DL + 0.8 WL(+Z)	753.55	-151.13	-25.30	83.73	33.58	5.62	122.93	111.69	55.76	44.52	79.73	117.31	50.14	89.35	78.11	
115 1.2 DL + 0.8 WL(-Z)	753.55	-151.13	-25.30	83.73	33.58	5.62	122.93	111.69	55.76	44.52	79.73	117.31	50.14	89.35	78.11	
0	429.84	0.00	0.00	47.76	0.00	0.00	47.76	47.76	47.76	47.76	4.56	47.76	47.76	47.76	47.76	
0	429.84	0.00	0.00	47.76	0.00	0.00	47.76	47.76	47.76	47.76	4.56	47.76	47.76	47.76	47.76	
0	429.84	0.00	0.00	47.76	0.00	0.00	47.76	47.76	47.76	47.76	4.56	47.76	47.76	47.76	47.76	
Max :	1441	0	101													
Min :	430	-261	-70													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 404.80 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 414.80 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 416.80 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 426.80 kN/m^2

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Sample calculation of values of base pressure:

Load case : 02 1 (DL+TL1) + 1 L

Maximum pressure, p_1 (pmax) = 127.41+43.60+4.29= 175.30 kN/m^2

p_2= 127.41+43.60-4.29= 166.73 kN/m^2

p_3= 127.41-(43.60+4.29)= 79.52 kN/m^2

p_4= 127.41-43.60+4.29= 88.09 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 448.00-2.4*18= 404.80 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (448.00+166.73)/2= 171.01 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (79.52+88.09)/2= 83.81 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (448.00+88.09)/2= 131.70 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (79.52+166.73)/2= 123.12 kN/m^2

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DESIGN OF FOOTING TYPE F9

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.52)+(0.6+2*0.5*0.52))*2

= 4.48 m

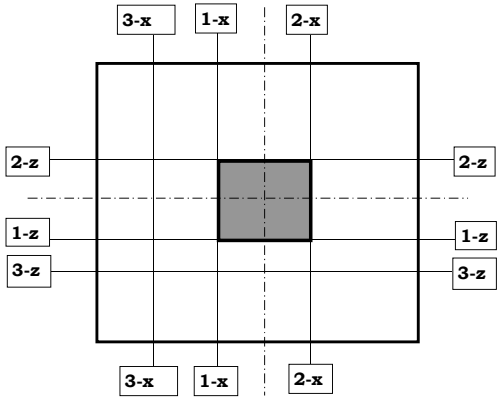
Punching Area= ((0.6+2*0.5*0.52)*(0.6+2*0.5*0.52))

= 1.25 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.6*24+1.8*18)

= 46.80 kN/m^2

Area of footing, A_f= 9.00 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions							
101 1.4 DL	108.73	92.15	123.66	60.52	16.72	56.75	415.66
102 1.2 (DL+TL1) + 1.6 LL	171.73	148.57	192.59	112.19	51.03	104.63	878.46
103 1.2 (DL+TL2) + 1.6 LL	171.73	148.57	192.59	112.19	51.03	104.63	878.46
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	118.70	105.46	130.60	64.47	29.54	60.11	505.92
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	126.16	105.37	144.88	77.11	22.20	72.28	534.46
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	103.28	87.84	117.18	55.49	14.72	52.06	377.88
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	141.58	122.99	158.30	86.08	37.02	80.34	662.50
108 0.9 DL + 1.6 WL(+X)	83.22	76.34	89.41	32.83	14.66	30.62	255.60
109 0.9 DL + 1.6 WL(-X)	90.69	76.24	103.69	45.47	7.33	42.79	284.14
110 0.9 DL + 1.6 WL(+Z)	67.81	58.71	75.99	23.86	-0.16	22.57	127.56
111 0.9 DL + 1.6 WL(-Z)	106.11	93.87	117.11	54.44	22.15	50.85	412.18
112 1.2 DL + 0.8 WL(+X)	98.16	85.83	109.25	48.81	16.27	45.69	350.21
113 1.2 DL + 0.8 WL(-X)	101.89	85.78	116.39	55.13	12.60	51.78	364.48
114 1.2 DL + 0.8 WL(+Z)	90.44	77.01	102.53	44.32	8.85	41.66	286.19
115 1.2 DL + 0.8 WL(-Z)	90.44	77.01	102.53	44.32	8.85	41.66	286.19
0	47.76	47.76	47.76	0.69	0.69	0.63	7.44
0	47.76	47.76	47.76	0.69	0.69	0.63	7.44
0	47.76	47.76	47.76	0.69	0.69	0.63	7.44

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1x} = 141.91-(141.91-58.97)*1.2/3 = 108.73 kN/m² Overhang = 1.2 m

p_{2x} = 58.97+(141.91-58.97)*1.2/3 = 92.15 kN/m² Effective depth = 0.54 m

p_{3x} = 141.91-(141.91-58.97)*(1.2-0.54)/3 = 123.66 kN/m²

BM_{1x} = (108.73-46.8)*1.2*1.2/2+0.5*(141.91-108.73)*1.2*1.2*0.67 = 60.52 kN-m/m

BM_{2x} = (92.15-46.8)*1.2*1.2/2-0.5*(-58.97+92.15)*1.2*1.2*0.67 = 16.72 kN-m/m

SF_{3x} = (123.66-46.8)*(1.2-0.54)+0.5*(141.91-123.66)*(1.2-0.54) = 56.75 kN/m

$S.F_{(punching)}$ = (9-1.25)*(100.43-46.8) = 415.66 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	112.19	0.38	0.51	0.001	665.36	972.00	20ϕT	323.21	300	200	1570.80
At Top	-0.16	0.00	0.00	0.000	0.91	972.00	20ϕT	323.21	300	200	1570.80

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 104.63 kN/m

Shear stress , τ_v= (104.63*1000)/(1000*540)

= 0.19 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 878.46 kN

Strength of Concrete in shear Vc1 = 7156.09 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 5260.58 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 4770.73 kN **Table 22.6.5.2_ (c)**

Design Vc= 4770.73 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (878.46*1000)/(4480*540)

= 0.36 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions						
101 1.4 DL	101.26	99.61	102.69	40.80	36.43	38.64
102 1.2 (DL+TL1) + 1.6 LL	161.29	159.01	163.26	84.62	78.60	80.06
103 1.2 (DL+TL2) + 1.6 LL	161.29	159.01	163.26	84.62	78.60	80.06
104 1.2 DL+ 1.6 WL(+X) + 0.5 LL	112.82	111.34	114.11	48.97	45.03	46.35
105 1.2 DL+ 1.6 WL(-X) + 0.5 LL	116.51	115.03	117.80	51.63	47.69	48.86
106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL	98.38	92.74	103.27	42.58	27.63	40.58
107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL	136.79	127.78	144.60	73.49	49.61	69.97
108 0.9 DL + 1.6 WL(+X)	80.21	79.35	80.97	24.90	22.60	23.57
109 0.9 DL + 1.6 WL(-X)	83.90	83.03	84.65	27.55	25.25	26.07
110 0.9 DL + 1.6 WL(+Z)	66.39	60.13	71.82	20.15	3.55	19.43
111 0.9 DL + 1.6 WL(-Z)	104.18	95.79	111.45	49.40	27.18	47.19
112 1.2 DL + 0.8 WL(+X)	92.65	91.34	93.80	34.29	30.79	32.47
113 1.2 DL + 0.8 WL(-X)	94.49	93.18	95.64	35.61	32.12	33.72
114 1.2 DL + 0.8 WL(+Z)	84.85	82.61	86.80	29.57	23.61	28.07
115 1.2 DL + 0.8 WL(-Z)	84.85	82.61	86.80	29.57	23.61	28.07
0	47.76	47.76	47.76	0.69	0.69	0.65
0	47.76	47.76	47.76	0.69	0.69	0.65
0	47.76	47.76	47.76	0.69	0.69	0.65

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Sample calculation of BM & SF

Load case : 101 1.4 DL

p_{1z} = 104.56-(104.56-96.31)*1.2/3= 101.26 kN/m^2

p_{2z} = 96.31+(104.56-96.31)*1.2/3= 99.61 kN/m^2

p_{3z} = 104.56-(104.56-96.31)*(1.2-0.52)/3= 102.69 kN/m^2

Overhang = 1.2 m

Effective depth = 0.52 m

BM_{1z} = (101.26-46.8)*1.2*1.2/2+0.5*(104.56-101.26)*1.2*1.2*0.67 = 40.80 kN-m/m

BM_{2z} = (99.61-46.8)*1.2*1.2/2-0.5*(-96.31+99.61)*1.2*1.2*0.67 = 36.43 kN-m/m

SF_{3z} = (102.69-46.8)*(1.2-0.52)+0.5*(104.56-102.69)*(1.2-0.52) = 38.64 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	84.62	0.313	0.417	0.0010	540.30	936.00	20ϕT	335.64	300	200	1570.8
Top	0.00	0.000	0.000	0.0000	0.00	936.00	20ϕT	335.64	300	200	1570.8

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 80.06 kN/m

Shear stress , τ_v= (80.06*1000)/(1000*520)

= 0.15 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **20ϕT** **200** c/c

At Top **20ϕT** **200** c/c

Along Z Direction :

At Bottom **20ϕT** **200** c/c

At Top **20ϕT** **200** c/c

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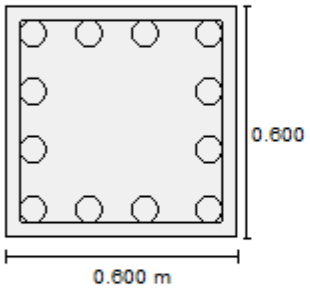
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5.5 Design of column C1, C2, C6 & C7 :

Geometry | Property | Loading | Shear Bending | Deflection | Concrete Design

Beam no. = 25 Design code : ACI-11



Design Load

Load	1
Location	END
Pu(Kns)	135.48
Mz(Kns-Mt)	5.6
My(Kns-Mt)	2.39

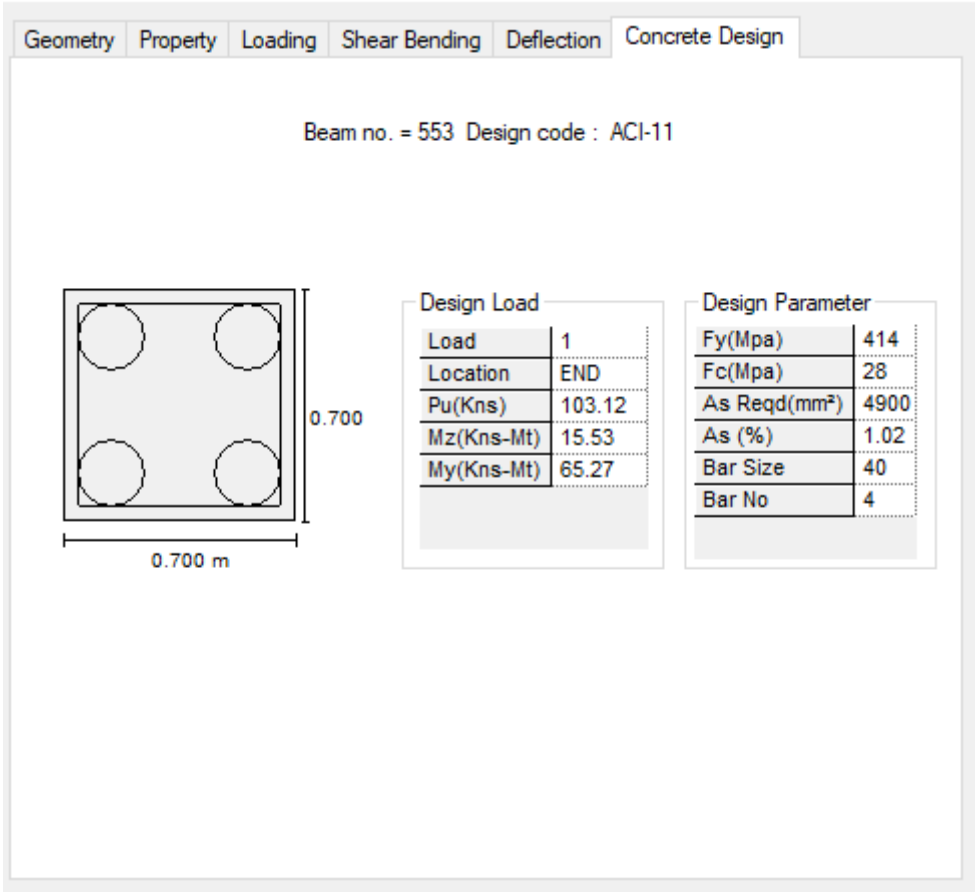
Design Parameter

Fy(Mpa)	414
Fc(Mpa)	28
As Reqd(mm²)	3600
As (%)	1.04
Bar Size	20
Bar No	12

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5.6 Design of column C8 & C9:



Provided 12 nos 25mm dia bar. Hence, Ast, provided = 5880 mm2

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REFERENCES:

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) clause 4.1.

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ATTACHMENT 1 STAAD INPUT FOR PLATFORM SUPER STRUCTURE:

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08-Apr-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
1 0 0 0; 2 5.4 0 0; 3 10.1 0 0; 4 15.1 0 0; 5 0 0 -4.5; 6 5.4 0 -4.5;
7 10.1 0 -4.5; 8 15.1 0 -4.5; 9 0 0 -8.5; 10 5.4 0 -8.5; 11 10.1 0 -8.5;
12 15.1 0 -8.5; 13 0 0 -15.25; 14 5.4 0 -15.25; 15 9.013 0 -15.25;
16 0 0 -18.3; 17 5.4 0 -18.3; 18 5.213 0 -19.23; 19 9.013 0 -19.23;
20 5.213 0 -24.53; 21 9.013 0 -24.53; 26 10.1 2.67 0; 27 15.1 2.67 0;
28 0 5.5 0; 29 5.4 5.5 0; 30 10.1 5.5 0; 31 15.1 5.5 0; 32 8 2.67 0;
33 10.1 8.57 0; 34 15.1 8.57 0; 35 8.55 8.57 0; 36 10.1 7.02 0; 37 0 8.61 0;
38 5.4 8.61 0; 39 0 11.72 0; 40 5.4 11.72 0; 41 0 14.62 0; 42 5.4 14.62 0;
43 2.7 14.62 0; 44 7 11.72 0; 45 5.4 10.22 0; 46 0 5.5 -4.5; 47 5.4 5.5 -4.5;
48 10.1 2.67 -4.5; 49 15.1 2.67 -4.5; 50 10.1 5.5 -4.5; 51 15.1 5.5 -4.5;
52 0 11.72 -4.5; 53 5.4 11.72 -4.5; 54 0 14.62 -4.5; 55 5.4 14.62 -4.5;
56 10.1 8.57 -4.5; 57 15.1 8.57 -4.5; 58 10.1 10.82 -4.5; 59 15.1 10.82 -4.5;
60 12.6 2.67 -4.5; 61 12.6 5.5 -4.5; 62 12.6 8.57 -4.5; 63 12.6 10.82 -4.5;
64 0 5.5 -8.5; 65 5.4 5.5 -8.5; 66 10.1 2.67 -8.5; 67 15.1 2.67 -8.5;
68 10.1 5.5 -8.5; 69 15.1 5.5 -8.5; 71 0 2.85 -8.5; 72 5.4 2.85 -8.5;
73 5.4 3.955 -8.5; 74 10.1 3.955 -8.5; 75 10.1 10.82 -8.5; 76 15.1 10.82 -8.5;
77 0 2.85 -15.25; 78 5.4 2.85 -15.25; 79 0 5.5 -15.25; 80 5.4 5.5 -15.25;
81 5.4 3.955 -15.25; 83 9.013 3.955 -15.25; 84 0 2.85 -18.3;
86 9.013 3.955 -19.23; 87 5.213 3.955 -19.23; 88 9.013 3.955 -24.53;
89 5.213 3.955 -24.53; 90 0 8.61 -4.5; 91 0 14.62 -2.155; 92 0 14.62 1.6;
93 0 13.07 0; 94 0 3.955 -4.5; 95 0 3.955 -8.5; 96 5.4 11.72 1.6;
97 5.4 3.955 -4.5; 98 5.4 1.053 -18.3; 99 5.4 1.053 -15.25; 100 5.4 5.5 -17;
101 6.167 5.5 -15.25; 102 5.4 4.733 -15.25; 103 10.1 8.57 1.55;
104 15.1 2.67 1.55; 105 15.1 1.12 0; 144 10.1 4.93 0; 145 10.1 4.93 1.55;
180 10.1 5.5 -5.25; 195 9.85 0 -25; 196 11.05 0 -25; 197 9.85 0 -22.85;
198 11.05 0 -22.85; 199 9.85 1.2 -25; 200 11.05 1.2 -25; 201 9.85 1.2 -22.85;
202 11.05 1.2 -22.85; 203 11.05 1.2 -23.925; 204 9.85 1.2 -23.925;
225 10.1 2.67 -7.35; 226 11.75 2.67 -7.35; 227 13.85 2.67 -7.35;
228 15.1 2.67 -7.35; 229 10.1 2.67 -5.25; 230 11.75 2.67 -5.25;
232 13.85 2.67 -5.25; 233 15.1 2.67 -5.25; 234 10.1 2.67 -6.3;
235 11.75 2.67 -6.3; 236 13.85 2.67 -6.3; 237 15.1 2.67 -6.3;
238 10.1 2.67 -1.1; 239 11.75 2.67 -1.1; 240 13.85 2.67 -1.1;
241 15.1 2.67 -1.1; 242 10.1 2.67 -3.3; 243 11.75 2.67 -3.3;
245 13.85 2.67 -3.3; 246 15.1 2.67 -3.3; 247 10.1 2.67 -2.25;
248 11.75 2.67 -2.25; 249 13.85 2.67 -2.25; 250 15.1 2.67 -2.25;
251 12.35 2.67 -7.35; 252 12.35 2.67 -5.25; 253 13.25 2.67 -7.35;
254 13.25 2.67 -5.25; 255 11.75 2.67 -6.75; 256 13.85 2.67 -6.75;
257 11.75 2.67 -5.85; 258 13.85 2.67 -5.85; 259 11.75 2.67 -1.7;
260 13.85 2.67 -1.7; 261 12.35 2.67 -1.1; 262 13.25 2.67 -1.1;
263 11.75 2.67 -2.6; 264 13.85 2.67 -2.6; 265 12.45 2.67 -3.3;
266 13.15 2.67 -3.3; 267 0 2.85 -10.63; 268 5.4 2.85 -10.63; 269 0 2.85 -14.46;
270 5.4 2.85 -14.46; 271 0.773 2.85 -10.63; 272 0.773 2.85 -14.46;
273 4.66 2.85 -10.63; 274 4.66 2.85 -14.46; 275 0.773 2.85 -11.88;

276 4.66 2.85 -11.88; 277 0.773 2.85 -13.21; 278 4.66 2.85 -13.21;
279 2.023 2.85 -10.63; 280 2.023 2.85 -14.46; 281 3.41 2.85 -10.63;
282 3.41 2.85 -14.46; 289 7.674 5.5 0; 290 7.674 5.5 -4.5; 292 7.674 5.5 -8.5;
293 7 5.5 0; 294 7 5.5 -4.5; 296 7 5.5 -8.5; 297 3.875 5.5 0;
298 3.875 5.5 -4.5; 300 3.875 5.5 -8.5; 301 3.875 5.5 -15.25; 302 1.525 5.5 0;
303 1.525 5.5 -4.5; 304 1.525 5.5 -8.5; 305 1.525 5.5 -15.25;
314 0 5.5 -11.875; 315 5.4 5.5 -11.875; 356 6.167 5.5 -8.5; 362 10.1 5.5 -7.35;
363 11.75 5.5 -7.35; 364 13.85 5.5 -7.35; 365 15.1 5.5 -7.35;
367 10.1 5.5 -6.5; 368 11.75 5.5 -6.5; 369 13.85 5.5 -6.5; 370 15.1 5.5 -6.5;
373 11.75 5.5 -5.25; 374 13.85 5.5 -5.25; 375 15.1 5.5 -5.25;
376 10.1 5.5 -3.3; 377 11.75 5.5 -3.3; 379 13.85 5.5 -3.3; 380 15.1 5.5 -3.3;
382 10.1 5.5 -2.25; 383 11.75 5.5 -2.25; 384 13.85 5.5 -2.15;
385 15.1 5.5 -2.15; 387 10.1 5.5 -1.1; 388 11.75 5.5 -1.1; 389 13.85 5.5 -1.1;
390 15.1 5.5 -1.1; 392 11.75 5.5 -6.75; 393 12.35 5.5 -7.35;
394 13.25 5.5 -7.35; 395 13.85 5.5 -6.75; 396 13.85 5.5 -5.85;
397 13.25 5.5 -5.25; 398 11.75 5.5 -5.85; 399 12.35 5.5 -5.25;
400 11.75 5.5 -2.6; 401 12.45 5.5 -3.3; 402 13.15 5.5 -3.3; 403 13.85 5.5 -2.6;
404 11.75 5.5 -1.7; 405 12.35 5.5 -1.1; 406 13.25 5.5 -1.1; 407 13.85 5.5 -1.7;
414 10.1 8.57 -3.3; 415 11.75 8.57 -3.3; 417 13.85 8.57 -3.3;
418 15.1 8.57 -3.3; 419 10.1 8.57 -2.15; 420 11.75 8.57 -2.15;
421 13.85 8.57 -2.15; 422 15.1 8.57 -2.15; 423 10.1 8.57 -1.1;
424 11.75 8.57 -1.1; 425 13.85 8.57 -1.1; 426 15.1 8.57 -1.1;
437 11.75 8.57 -2.6; 438 12.45 8.57 -3.3; 439 13.15 8.57 -3.3;
440 13.85 8.57 -2.6; 441 11.75 8.57 -1.7; 442 12.35 8.57 -1.1;
443 13.25 8.57 -1.1; 444 13.85 8.57 -1.7; 446 13.85 10.82 -5.25;
448 11.75 10.82 -5.25; 449 11.75 10.82 -7.35; 451 10.1 10.82 -7.35;
452 13.85 10.82 -7.35; 454 15.1 10.82 -7.35; 455 12.35 10.82 -7.35;
456 11.75 10.82 -6.5; 457 11.75 10.82 -6.75; 458 10.1 10.82 -6.5;
459 13.85 10.82 -6.5; 460 13.85 10.82 -6.75; 461 15.1 10.82 -6.5;
462 11.75 10.82 -5.85; 463 10.1 10.82 -5.25; 464 13.85 10.82 -5.85;
465 15.1 10.82 -5.25; 466 12.35 10.82 -5.25; 467 13.25 10.82 -7.35;
468 13.25 10.82 -5.25; 469 0.83 11.72 0; 470 0.83 11.72 -4.5; 471 4.57 11.72 0;
472 4.57 11.72 -4.5; 473 0.83 11.72 -4.02; 474 4.57 11.72 -4.02;
475 0.83 11.72 -2.74; 476 4.57 11.72 -2.74; 477 0.83 11.72 -1.521;
478 4.57 11.72 -1.521; 479 2.251 11.72 0; 480 3.149 11.72 0;
481 2.01 11.72 -4.02; 482 3.39 11.72 -4.02; 497 5.4 14.62 -2.155;
506 2.7 5.5 0; 507 12.6 2.67 0; 508 12.6 8.57 0; 509 0 10.22 -4.5;
512 5.4 10.22 -4.5; 513 0 2.67 -4.5; 514 5.4 2.67 -4.5; 515 2.7 3.955 -4.5;
516 2.7 5.5 -4.5; 523 1.35 10.22 -4.5; 524 4.05 10.22 -4.5; 525 2.7 11.72 -4.5;
526 2.7 14.62 -4.5; 527 2.7 2.85 -8.5; 528 2.7 5.5 -8.5; 529 10.1 8.57 -8.5;
530 15.1 8.57 -8.5; 531 12.6 2.67 -8.5; 532 12.6 5.5 -8.5; 533 12.6 10.82 -8.5;
534 12.6 8.57 -8.5; 535 2.7 2.85 -15.25; 536 5.213 3.955 -21.88;
537 9.013 1.9775 -15.25; 538 9.013 1.9775 -19.23; 539 9.013 3.955 -17.24;
540 9.013 1.9775 -17.24; 541 0 2.67 0; 542 5.4 2.67 0; 543 2.7 2.67 0;
544 0 2.67 -2.25; 545 0 5.5 -2.25; 546 0 2.85 -11.875; 547 5.4 1.4 -2.25;
548 5.4 5.5 -2.25; 553 5.4 11.72 -2.25; 555 10.1 8.57 -6.5; 560 15.1 8.57 -6.5;
564 5.4 2.85 -11.875; 566 5.4 2.85 -9.565; 569 0.773 2.85 -15.25;
570 4.66 2.85 -15.25; 571 0.773 2.85 -8.5; 573 0 2.85 -14.855;
574 5.4 2.85 -14.855; 577 1.398 2.85 -15.25; 578 4.035 2.85 -15.25;
579 0.3865 2.85 -15.25; 580 0.3865 2.85 -14.46; 581 5.03 2.85 -14.46;

582 5.03 2.85 -15.25; 583 0.773 2.85 -14.855; 584 4.66 2.85 -14.855;
587 0 2.85 -11.2525; 588 0 2.85 -13.835; 589 0 2.85 -12.5425;
590 0 2.85 -9.0325; 591 0 2.85 -10.0975; 592 0.773 2.85 -9.565;
593 10.7 2.67 -4.5; 594 10.7 2.67 -5.25; 595 14.5 2.67 -4.5;
596 14.5 2.67 -5.25; 597 10.1 2.67 -4.875; 598 15.1 2.67 -4.875;
599 10.7 5.5 -4.5; 600 10.7 5.5 -5.25; 601 14.5 5.5 -4.5; 602 14.5 5.5 -5.25;
603 10.1 5.5 -4.875; 604 15.1 5.5 -4.875; 605 11 8.57 -3.3; 606 11 8.57 -4.5;
607 14.5 8.57 -3.3; 608 14.2 8.57 -4.5; 609 10.1 8.57 -3.9; 610 15.1 8.57 -3.9;
611 10.7 10.82 -4.5; 612 10.7 10.82 -5.25; 613 14.5 10.82 -4.5;
614 14.5 10.82 -5.25; 615 10.1 10.82 -4.875; 616 15.1 10.82 -4.875;
617 7 11.72 -4.5; 618 6.2 11.72 -4.5; 619 6.2 11.72 0; 620 0 11.72 -4.02;
621 0 11.72 -0.43; 622 0.83 11.72 -0.43; 623 0.415 11.72 0;
624 0.415 11.72 -4.5; 627 0 11.72 -0.9755; 628 0 11.72 -3.43;
629 0 11.72 -2.1555; 669 0 8.61 -2.25; 670 0 11.72 -2.25; 671 2.7 8.61 0;
672 2.7 11.72 0; 679 2.7 5.5 -15.25; 680 4.625 5.5 0; 681 4.625 5.5 -4.5;
682 4.625 5.5 -8.5; 683 4.625 5.5 -15.25; 684 0.775 5.5 0; 685 0.775 5.5 -4.5;
686 0.775 5.5 -8.5; 687 0.775 5.5 -15.25; 688 3.875 5.5 -9.34375;
690 3.875 5.5 -11.0313; 691 3.875 5.5 -11.875; 692 3.875 5.5 -12.7188;
694 3.875 5.5 -14.4063; 695 1.525 5.5 -9.34375; 697 1.525 5.5 -11.0313;
698 1.525 5.5 -11.875; 699 1.525 5.5 -12.7188; 701 1.525 5.5 -14.4063;
703 4.625 5.5 -10.1875; 705 4.625 5.5 -11.875; 707 4.625 5.5 -13.5625;
710 0.775 5.5 -10.1875; 712 0.775 5.5 -11.875; 714 0.775 5.5 -13.5625;
716 2.7 5.5 -11.875; 717 6.167 5.5 -11.875; 718 3.875 5.5 -1.125;
720 3.875 5.5 -3.375; 721 3.875 5.5 -5.5; 723 3.875 5.5 -7.5;
724 1.525 5.5 -1.125; 726 1.525 5.5 -3.375; 727 1.525 5.5 -5.5;
729 1.525 5.5 -7.5; 731 4.625 5.5 -2.25; 734 4.625 5.5 -6.5;
737 0.775 5.5 -2.25; 740 0.775 5.5 -6.5; 741 8.387 5.5 0; 742 8.387 5.5 -4.5;
743 8.387 5.5 -8.5; 744 6.167 5.5 -4.5; 745 6.167 5.5 0; 746 5.4 5.5 -6.5;
747 7 5.5 -2.25; 748 7 5.5 -6.5; 749 6.167 5.5 -6.5; 750 6.167 5.5 -2.25;
752 7.674 5.5 -6.5; 754 8.387 5.5 -7.5; 756 8.387 5.5 -5.5;
758 7.674 5.5 -2.25; 760 8.387 5.5 -3.375; 762 8.387 5.5 -1.125;
763 8.387 5.5 -6.5; 764 8.387 5.5 -2.25; 765 0 5.5 -14.4063;
766 5.4 5.5 -14.4063; 767 2.7 5.5 -14.4063; 768 4.625 5.5 -14.4063;
769 0.775 5.5 -14.4063; 770 1.525 5.5 -1.7; 771 2.7 5.5 -1.7;
772 3.875 5.5 -1.7; 773 1.525 5.5 -13.038; 774 2.7 5.5 -13.038;
775 3.875 5.5 -13.038; 776 1.35 10.165 0; 777 4.05 10.165 0; 778 12.6 4.085 0;
779 2.7 4.175 -15.25; 780 7.75 5.5 -4.5; 781 5.4 7.9 0; 782 5.4 7.9 -4.5;
783 5.4 9.4 -4.5; 784 5.4 9.4 0; 785 5.4 7.9 -3.375; 786 5.4 9.4 -2.25;
787 5.4 7.9 -1.125; 789 0.5 14.62 0; 791 4.4 14.62 0; 793 0.5 14.62 -4.5;
794 0 14.62 -1.16667; 799 5.4 14.62 -1.16667; 800 4.4 14.62 -4.5;
802 1 14.62 0; 803 1 14.62 -4.5; 804 0 14.62 -2.33333; 805 0 14.62 -3.5;
806 5.4 14.62 -2.33333; 807 5.4 14.62 -3.5; 819 0 14.62 -3.8;
820 5.4 14.62 -3.8; 843 4.9 14.62 0; 844 4.9 14.62 -4.5; 845 3.55 14.62 0;
846 3.55 14.62 -4.5; 847 1.85 14.62 0; 848 1.85 14.62 -4.5;
851 3.55 14.62 -3.5; 852 1.85 14.62 -3.5; 855 3.55 14.62 1.6;
856 1.85 14.62 1.6; 858 5.4 14.62 1.6; 859 5.4 14.62 0.9; 860 0 14.62 0.9;
861 3.55 14.62 0.9; 862 1.85 14.62 0.9; 863 3.55 14.62 -1.16667;
864 3.55 14.62 -2.33333; 865 1.85 14.62 -1.16667; 866 1.85 14.62 -2.33333;
867 0 11.72 1.6; 871 0 14.62 2.4; 872 5.4 14.62 2.4; 873 2.7 14.62 1.6;
874 3.55 14.92 -3.5; 875 1.85 14.92 -3.5; 876 3.55 14.92 0.9;

877 1.85 14.92 0.9; 878 3.55 14.92 -1.16667; 879 1.85 14.92 -1.16667;
880 1.85 14.92 0.486666; 881 1.85 14.92 0.073332; 882 1.85 14.92 -0.340002;
883 1.85 14.92 -0.753336; 884 3.55 14.92 0.486666; 885 3.55 14.92 0.073332;
886 3.55 14.92 -0.340002; 887 3.55 14.92 -0.753336; 888 3.55 14.92 -1.63334;
889 3.55 14.92 -2.1; 890 3.55 14.92 -2.56667; 891 3.55 14.92 -3.03333;
892 1.85 14.92 -1.63334; 893 1.85 14.92 -2.1; 894 1.85 14.92 -2.56667;
895 1.85 14.92 -3.03333; 899 1.225 3.955 -4.5; 900 4.175 3.955 -4.5;
901 5.4 1.4 0; 902 5.4 1.4 -4.5; 903 5.4 1.4 -3.375; 904 5.4 1.4 -1.125;
905 5.4 4.085 0; 906 5.4 4.085 -1.125; 907 5.4 4.085 -4.5;
908 5.4 4.085 -3.375; 909 5.4 11.72 0.775; 910 0 11.72 0.8;
911 9.013 1.9775 -21.88; 913 13.85 12.6 -5.25; 920 13.85 12.6 -7.35;
921 15.1 12.6 -7.35; 922 15.1 12.6 -5.25; 923 13.85 12.6 -5.67;
924 13.85 12.6 -6.09; 925 13.85 12.6 -6.51; 926 13.85 12.6 -6.93;
927 13.05 12.6 -5.25; 928 13.05 12.6 -7.35; 929 13.85 12.22 -5.25;
930 13.85 12.22 -7.35; 931 13.55 10.82 -4.5; 932 11.75 10.82 -4.5;
935 11.75 10.82 -3.3; 936 14.5 10.82 -3.3; 937 14.5 10.82 -3.6;
938 14.5 10.82 -3.9; 939 14.5 10.82 -4.2; 940 11.75 10.82 -3.6;
941 11.75 10.82 -3.9; 942 11.75 10.82 -4.2; 943 7 5.5 -0.45; 944 7 5.5 -0.9;
945 7 5.5 -1.35; 946 7 5.5 -1.8; 947 7 5.5 -4.9; 948 7 5.5 -5.3;
949 7 5.5 -5.7; 950 7 5.5 -6.1; 951 6.167 5.5 -8.1; 952 6.167 5.5 -7.7;
953 6.167 5.5 -7.3; 954 6.167 5.5 -6.9; 955 6.167 5.5 -4.05;
956 6.167 5.5 -3.6; 957 6.167 5.5 -3.15; 958 6.167 5.5 -2.7; 959 7 5.5 -2.7;
960 7 5.5 -3.15; 961 7 5.5 -3.6; 962 7 5.5 -4.05; 963 7 5.5 -6.9;
964 7 5.5 -7.3; 965 7 5.5 -7.7; 966 7 5.5 -8.1; 967 6.167 5.5 -6.1;
968 6.167 5.5 -5.7; 969 6.167 5.5 -5.3; 970 6.167 5.5 -4.9; 971 6.167 5.5 -1.8;
972 6.167 5.5 -1.35; 973 6.167 5.5 -0.9; 974 6.167 5.5 -0.45;
975 7.674 5.5 -7.3; 976 7.674 5.5 -6.9; 977 7.674 5.5 -5.7; 978 7.674 5.5 -5.3;
979 7.674 5.5 -3.6; 980 7.674 5.5 -4.05; 981 7.674 5.5 -2.7;
982 7.674 5.5 -1.35; 983 7.674 5.5 -0.9; 984 10.7 5.5 -3.3; 985 12.6 5.5 -3.3;
986 14.5 5.5 -3.3; 987 10.1 5.5 -3.9; 988 15.1 5.5 -3.9; 989 7.22467 5.5 -7.3;
990 7.44933 5.5 -7.3; 991 7.22467 5.5 -6.9; 992 7.44933 5.5 -6.9;
993 7.22467 5.5 -5.7; 994 7.44933 5.5 -5.7; 995 7.22467 5.5 -5.3;
996 7.44933 5.5 -5.3; 997 7.22467 5.5 -3.6; 998 7.44933 5.5 -3.6;
999 7.22467 5.5 -4.05; 1000 7.44933 5.5 -4.05; 1001 7.22467 5.5 -2.7;
1002 7.44933 5.5 -2.7; 1003 7.22467 5.5 -2.25; 1004 7.44933 5.5 -2.25;
1005 7.22467 5.5 -1.35; 1006 7.44933 5.5 -1.35; 1007 7.22467 5.5 -0.9;
1008 7.44933 5.5 -0.9; 1009 7 5.5 -2.475; 1010 7 5.5 -1.125; 1011 7 5.5 -5.5;
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_FLOOR-1 285 TO 288 291 292
_FLOOR-2_1 91 319 321 323 325 TO 332 334 TO 341 363 TO 365 367 TO 374 991 994
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383 TO 386 909 923
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1485 TO 1488 1515 1517 TO 1540 1542 TO 1549 1567 TO 1570 1600 1603 -
1799 TO 1801
_NEWTOPFLOOR-2 28 140 715 718 728 729 1063 1179 1408 1413 1415 1480 1486 1488 -
1516 1518 1521 1522 1541 1564 1566 1631 TO 1633 1799 TO 1801

_NEWPLATFORM 66 276 1638 TO 1640 1644 1649 TO 1656 1661 1663 1668 TO 1673
_VALVEMAINTPLAT 879 1616 1788 1789 1793 1794 1796 1797
END GROUP DEFINITION
START USER TABLE
TABLE 1
UNIT METER MTON
GENERAL
IPE300-HLFCTIPE300
0.008286 0.3 0.007 0.229 0.011 8.63611e-05 3.75143e-05 2.84755e-07 -
0.000575741 0.00032745 0.00244308 0.00126791 0.000695518 0.000447615 -
1.94458e-07 0.3
PROFILE_POINTS
 -0.114435 0.15 -0.114435 0.139 -0.0434348 0.139 -0.0434348 -
-0.139 -0.114435 -0.139 -0.114435 -0.15 0.0355652 -0.15 0.0355652 -0.139 -
-0.0354348 -0.139 -0.0354348 -0.004 0.103565 -0.004 0.103565 -0.075 0.114565 -
-0.075 0.114565 0.075 0.103565 0.075 0.103565 0.004 -0.0354348 0.004 -
-0.0354348 0.139 0.0355652 0.139 0.0355652 0.15 -0.114435 0.15
END
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+07
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-05
DAMP 0.03
TYPE STEEL
STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
102 105 106 108 109 111 112 281 TO 284 851 852 1524 TO 1526 1535 1536 1543 -
1545 1547 1549 1564 1566 TO 1570 1625 1626 1632 1633 1663 TO 1665 1786 1787 -
1800 1801 TABLE ST IPE240
1 TO 4 6 10 11 21 23 26 27 41 TO 46 53 TO 56 59 TO 62 75 TO 80 85 86 88 89 -
93 TO 96 98 99 101 117 127 130 131 142 145 149 156 164 166 215 217 270 271 -
637 639 641 643 TO 645 650 654 656 658 666 668 672 674 768 769 787 -
790 TO 792 885 898 912 1022 1025 1028 1029 1390 1404 1405 1562 1563 1614 -
1615 1618 1621 1806 1807 TABLE ST IPE300
25 73 74 778 779 788 795 796 801 811 819 845 846 848 TO 850 862 TO 865 873 -
874 895 926 927 1176 TO 1178 1180 1181 1377 1378 1385 1571 1572 1606 -
1798 TABLE LD L90X90X8 SP 0.01
289 290 293 TO 296 TABLE ST L75X75X8
7 14 15 19 20 39 47 50 57 58 63 TO 66 83 91 92 119 134 139 140 146 147 150 -
152 TO 155 157 158 160 TO 163 165 167 216 260 276 285 TO 288 291 292 316 -
319 321 323 325 TO 332 334 TO 349 351 353 355 356 358 TO 360 362 TO 386 407 -
408 TO 410 414 417 539 550 552 554 TO 556 560 TO 562 564 569 TO 575 577 579 -
580 TO 581 585 TO 588 590 591 594 TO 600 603 TO 626 631 632 636 646 TO 648 -
651 665 667 669 TO 671 673 675 676 683 TO 685 687 TO 690 692 693 695 TO 714 -
732 TO 735 738 891 903 906 909 915 916 923 985 990 991 993 994 996 997 1007 -
1009 1010 1012 1015 1023 1026 1037 1039 1040 1042 1045 1052 1055 -
1057 TO 1059 1069 1070 1072 1172 1175 1194 1195 1197 1200 1202 1204 1207 -

1209 1211 1227 1421 1600 1631 1637 TO 1644 1649 TO 1652 1657 1659 1661 1662 -
1739 1741 1743 TO 1745 TABLE ST IPE240
1368 1369 1371 1374 1376 TABLE ST IPE240
1229 1231 1234 1236 1238 1260 1263 1278 1281 1284 1287 1308 1309 1312 1313 -
1317 1318 1328 1330 1338 1340 1345 1346 1354 TO 1362 1373 1682 TO 1689 1698 -
1699 TO 1705 1769 TO 1771 1774 1775 TABLE ST IPE240
934 935 953 TO 956 963 TO 970 975 976 978 TO 981 984 987 TO 989 992 995 998 -
999 TO 1005 1008 1011 1013 1014 1016 TO 1021 1024 1027 1030 TO 1035 1038 1041 -
1043 1044 1046 TO 1051 1061 1062 1065 TO 1068 1074 TO 1079 1240 TO 1259 1261 -
1262 1264 1289 TO 1304 1319 TO 1324 1331 TO 1334 1341 TO 1344 1347 TO 1351 -
1363 TO 1366 1725 TO 1732 1736 TO 1738 1790 TO 1792 1795 1804 1805 -
1808 TO 1811 TABLE ST L65X65X6
16 22 40 148 151 159 168 218 899 900 1054 1379 1380 1382 1384 1636 -
1645 TO 1648 1658 1660 TABLE LD L65X65X6 SP 0.01
798 799 812 814 TO 816 1396 1398 1400 1401 1612 1613 TABLE LD L65X65X6 SP 0.01
67 TO 72 122 123 141 780 783 TO 786 821 822 824 825 828 832 TO 841 843 844 -
847 853 TO 859 868 869 871 872 880 881 883 884 896 897 901 902 904 905 907 -
908 910 911 913 914 917 TO 922 924 925 1170 1171 1173 1174 1381 1383 1391 -
1394 1395 1397 1399 1402 1403 1607 1619 1623 1627 TO 1630 1802 -
1803 TABLE LD L75X75X8 SP 0.01
876 877 929 930 1387 1388 TABLE LD L130X130X10 SP 0.01
87 100 395 398 TO 402 820 842 936 TO 938 940 TO 942 951 952 957 960 TO 962 -
986 TABLE ST IPE360
82 133 144 387 TO 394 396 397 403 TO 406 422 423 425 426 430 432 433 538 875 -
928 931 932 943 944 958 959 971 973 974 983 1307 1315 1316 1326 1336 1531 -
1532 1674 TO 1681 1690 TO 1697 1706 1708 1710 1712 1714 1716 1718 1721 1723 -
1766 TO 1768 1772 1773 TABLE ST IPE450
28 51 715 TO 722 724 TO 729 805 1060 1063 1064 1179 TABLE ST IPE400
723 730 731 TABLE ST IPE360
1408 1413 1415 1480 1486 1488 1516 1518 TABLE ST IPE400
MEMBER PROPERTY EUROPEAN
12 48 81 97 114 TO 116 135 TO 137 435 436 439 440 442 443 445 447 449 450 -
511 514 777 797 823 1196 1198 1199 1201 1203 1205 1206 1208 1210 1314 1352 -
1353 TABLE ST IPE300
90 104 110 113 132 143 TABLE ST IPE360
MEMBER PROPERTY EUROPEAN
437 438 441 444 446 448 1212 1214 TO 1216 1218 1219 1221 TO 1223 1225 1265 -
1267 1268 1270 1271 1273 1274 1276 1367 1370 1372 -
1375 TAPERED 0.3 0.01 0.3 0.22 0.016 0.22 0.016
1550 TO 1561 TABLE ST L65X65X6
MEMBER PROPERTY EUROPEAN
1521 1522 TAPERED 0.4 0.01 0.4 0.25 0.016 0.25 0.016
1573 TO 1598 TABLE ST IPE300
793 794 878 879 1610 1611 1616 1617 1620 1622 1624 TABLE ST UPN300
MEMBER PROPERTY FRENCH
34 1605 TABLE LD L100X100X8 SP 0.01
MEMBER PROPERTY EUROPEAN
1634 1635 TABLE LD L150X150X13 SP 0.008
13 49 421 428 429 1305 1306 1310 1311 1386 1406 1409 1410 1412 1414 1416 1417 -
1425 1426 1479 1485 1487 1515 1517 1528 1533 1534 1537 TO 1541 1599 -

1603 TABLE ST HE500A
MEMBER PROPERTY EUROPEAN
1519 1520 1542 1544 1546 1548 TAPERED 0.4 0.012 0.4 0.2 0.02 0.2 0.02
MEMBER PROPERTY EUROPEAN
1523 1527 1529 1530 TABLE ST IPE450
1799 TABLE ST UPN240
1653 TO 1656 1666 TO 1673 1707 1709 1711 1713 1715 1717 1719 1720 1722 1724 -
1733 TO 1735 1746 TO 1765 TABLE ST UPN200
1776 TO 1785 1788 1789 1793 1794 1796 1797 TABLE ST IPE200
118 866 867 1169 TABLE LD L110X110X10
MEMBER PROPERTY EUROPEAN
8 9 24 1389 UPTABLE 1 IPE300-HLFCTIPE300
CONSTANTS
BETA 90 MEMB 3 4 6 10 11 21 43 TO 46 59 TO 62 77 TO 80 89 93 94 166 215 217 -
281 TO 284 793 794 878 879 898 912 1573 TO 1576 1580 1582 1610 1611 1616 -
1617 1620 1622 1624 1663 TO 1665
MATERIAL STEEL ALL
SUPPORTS
1 TO 21 195 TO 198 PINNED
84 98 100 103 104 145 FIXED BUT FY FZ MX MY MZ
32 35 FIXED BUT FX FY MX MY MZ
MEMBER RELEASE
7 12 TO 14 19 25 28 47 TO 51 57 81 TO 83 87 90 91 97 100 104 114 TO 116 118 -
119 123 133 135 TO 137 139 143 144 152 TO 156 160 TO 164 316 319 323 329 -
330 TO 331 338 341 343 TO 345 351 359 362 371 TO 374 383 TO 386 389 392 395 -
398 407 TO 410 414 423 425 430 432 437 438 441 444 446 448 TO 450 511 514 -
539 552 561 564 569 TO 571 577 587 591 595 TO 597 605 608 611 614 617 620 -
623 626 632 639 647 651 654 667 670 673 676 684 685 692 696 698 699 705 708 -
711 714 717 720 723 732 TO 735 738 780 788 793 828 845 TO 847 862 866 878 -
896 899 913 940 TO 942 985 1057 1180 1181 1194 1195 1197 1200 1202 1204 1207 -
1209 1211 1308 1309 1312 1313 1357 TO 1362 1369 1371 1374 1376 1385 1387 -
1388 1406 1409 1412 1519 TO 1524 1528 1531 1532 1535 1536 1543 1545 1547 -
1549 TO 1561 1566 1569 1571 TO 1576 1580 1582 1625 1626 1633 1634 -
1636 TO 1648 1653 TO 1656 1658 1660 1663 TO 1665 1707 1709 1711 1713 1715 -
1717 1719 1720 1722 1724 1733 TO 1735 1776 TO 1780 1788 TO 1799 1804 1805 -
1808 TO 1811 START MY MZ
14 90 115 143 286 287 321 326 327 338 341 348 349 355 356 359 362 367 368 -
371 TO 374 379 380 383 TO 386 396 401 402 407 TO 410 550 554 555 561 564 -
587 591 600 603 605 607 608 611 614 615 617 619 620 623 626 647 651 658 665 -
667 669 TO 671 673 675 676 688 695 703 705 707 708 711 714 718 719 721 722 -
731 TO 735 738 811 832 845 848 854 863 867 891 900 914 943 944 959 TO 962 -
985 986 993 994 1009 1010 1025 1026 1028 1029 1040 1057 1058 1169 1176 1195 -
1198 1199 1201 1203 1218 1225 1263 1267 1270 1273 1276 1278 1281 1284 1287 -
1305 TO 1307 1314 1330 1340 1354 TO 1362 1368 1369 1371 1374 1376 TO 1378 -
1381 1385 1387 1388 1426 1487 1488 1521 1522 1526 1529 1531 TO 1533 1535 -
1536 1538 1540 1543 TO 1545 1547 TO 1561 1566 1569 1571 1572 1611 1617 1620 -
1632 1633 1635 1636 1638 1639 1644 TO 1648 1652 1655 1656 1658 1660 1663 -
1670 1673 1693 1697 1701 1705 1706 1716 1733 TO 1735 1743 TO 1745 1747 1749 -
1751 1753 1755 1757 1759 1761 1763 1765 1781 TO 1799 1802 TO 1805 -
1808 TO 1811 END MY MZ

20 END MZ
1631 END MZ
158 END MZ
216 END MZ
167 END MZ
39 147 START MZ
58 1052 START MZ
15 END MZ
150 END MZ
134 END MZ
134 START MZ
146 START MZ
146 END MZ
287 START MZ
286 START MZ
292 END MZ
285 START MZ
288 START MZ
291 END MZ
92 START MZ
683 END MZ
157 START MZ
1039 END MZ
165 START MZ
687 END MZ
1408 START MZ
1425 END MZ
132 END MZ
132 START MZ
855 END MZ
853 START MZ
MEMBER TRUSS
16 22 34 40 67 TO 74 122 141 148 159 168 218 289 290 293 TO 296 778 779 783 -
784 TO 786 795 796 798 799 812 814 TO 816 819 821 822 824 825 833 TO 841 843 -
844 849 850 856 TO 859 864 865 868 869 871 TO 874 876 877 880 881 883 884 -
895 897 901 902 904 905 907 908 910 911 917 TO 922 924 TO 927 929 930 934 -
935 953 TO 956 963 TO 970 975 976 978 TO 981 984 987 TO 989 992 995 -
998 TO 1005 1008 1011 1013 1014 1016 TO 1021 1024 1027 1030 TO 1035 1038 -
1041 1043 1044 1046 TO 1051 1054 1061 1062 1065 TO 1068 1074 TO 1079 1170 -
1171 1173 1174 1177 1178 1240 TO 1259 1261 1262 1264 1289 TO 1304 -
1319 TO 1324 1331 TO 1334 1341 TO 1344 1347 TO 1351 1363 TO 1366 1379 1380 -
1382 1383 1396 1398 1400 1401 1605 TO 1607 1612 1613 1619 1623 1627 TO 1630 -
1725 TO 1732 1736 TO 1738
*
LOAD 1 LOADTYPE Dead TITLE DEAD LOAD
SELFWEIGHT Y -1 LIST 1 TO 4 6 TO 16 19 TO 28 34 39 TO 51 53 TO 83 85 TO 102 -
104 TO 106 108 TO 119 122 123 127 130 TO 137 139 TO 168 215 TO 218 260 270 -
271 276 281 TO 296 316 319 321 323 325 TO 332 334 TO 349 351 353 355 356 -
358 TO 360 362 TO 410 414 417 421 TO 423 425 426 428 TO 430 432 433 -
435 TO 450 511 514 538 539 550 552 554 TO 556 560 TO 562 564 569 TO 575 577 -

579 TO 581 585 TO 588 590 591 594 TO 600 603 TO 626 631 632 636 637 639 641 -
643 TO 648 650 651 654 656 658 665 TO 676 683 TO 685 687 TO 690 692 693 695 -
696 TO 735 738 768 769 777 TO 780 783 TO 788 790 TO 799 801 805 811 812 814 -
815 TO 816 819 TO 825 828 832 TO 859 862 TO 869 871 TO 881 883 TO 885 891 -
895 TO 932 934 TO 938 940 TO 944 951 TO 971 973 TO 976 978 TO 981 -
983 TO 1005 1007 TO 1035 1037 TO 1052 1054 1055 1057 TO 1070 1072 -
1074 TO 1079 1169 TO 1181 1194 TO 1212 1214 TO 1216 1218 1219 1221 TO 1223 -
1225 1227 1229 1231 1234 1236 1238 1240 TO 1265 1267 1268 1270 1271 1273 -
1274 1276 1278 1281 1284 1287 1289 TO 1324 1326 1328 1330 TO 1334 1336 1338 -
1340 TO 1391 1394 TO 1406 1408 TO 1410 1412 TO 1417 1421 1425 1426 1479 1480 -
1485 TO 1488 1515 TO 1564 1566
SELFWEIGHT Y -1 LIST 1567 TO 1600 1603 1605 TO 1607 1610 TO 1633 1638 TO 1640
-
1644 1649 TO 1656 1661 1663 1668 TO 1673
***Grating Weight
FLOOR LOAD
_FLOOR-1 FLOAD -0.035 GY
_FLOOR-2_1 FLOAD -0.035 GY
_FLOOR-2_2 FLOAD -0.035 GY
_FLOOR-2_3 FLOAD -0.035 GY
_FLOOR-3_1 FLOAD -0.035 GY
_FLOOR-3_2 FLOAD -0.035 GY
_FLOOR-4_1 FLOAD -0.035 GY
_FLOOR-4_2 FLOAD -0.035 GY
_FLOOR-4_3 FLOAD -0.035 GY
_FLOOR-5_1 FLOAD -0.035 GY
_FLOOR-5_2 FLOAD -0.035 GY
_FLOOR-7_1 FLOAD -0.035 GY
_FLOOR-7_2 FLOAD -0.035 GY
_FLOOR-8_1 FLOAD -0.035 GY
_FLOOR-8_2 FLOAD -0.035 GY
_FLOOR-9_1 FLOAD -0.035 GY
_FLOOR-9_2 FLOAD -0.035 GY
MEMBER LOAD
*Hand Rail Weight
7 12 TO 14 19 28 39 51 57 58 65 66 83 87 91 92 97 100 114 TO 116 119 133 144 -
150 152 153 156 157 160 TO 165 270 271 276 316 319 321 326 328 334 336 340 -
342 347 349 355 387 388 390 391 414 421 426 428 433 435 440 442 447 449 511 -
538 539 554 572 580 590 599 632 637 643 645 650 683 685 687 697 700 715 716 -
718 719 728 729 738 777 805 820 842 875 903 909 915 916 923 928 931 932 936 -
937 TO 938 943 944 951 952 957 960 971 973 974 983 996 997 1015 1023 1026 -
1028 1029 1039 1042 1045 1052 1055 1058 1059 1063 1064 1069 1070 1072 1172 -
1175 1179 1196 1198 1203 1205 1210 1263 1305 1307 1311 1352 1408 TO 1410 -
1412 1413 1415 TO 1417 1421 1425 1426 1480 1486 1488 1516 1518 1523 1527 -
1529 1530 1537 1539 1567 TO 1569 1600 1603 1661 1662 1745 1799 TO 1800 -
1801 UNI GY -0.015
***Stair Weight
15 20 140 158 167 216 CON GY -0.25 0.075
15 20 158 167 216 CON GY -0.25 1.275
1631 CON GY -0.25 0.5

58 654 CON GY -0.25 0.225
654 CON GY -0.25 1.425
276 CON GY -0.25 0.825
134 146 147 CON GY -0.25 0.35
134 146 147 CON GY -0.25 1.55
134 146 CON GY -0.25 1.85
134 146 CON GY -0.25 3.05
285 287 288 CON GY -0.25 0
287 CON GY -0.25 1.2
291 292 CON GY -0.25 0.125
*Monorail Weight
110 113 CON GY -0.5 1.8235
90 104 CON GY -0.5 1.636
132 143 CON GY -0.5 1.992
*3660-TNK-003
399 400 403 404 CON GY -1.5
*3660-FAN-001
1216 1223 CON GY -1 0.319
1212 1219 CON GY -2 1.408
1270 1276 CON GY -1 0.4
1267 1273 CON GY -2.2 0.822
1367 1370 CON GY -1 0
*3660-HEX-001 to 005
JOINT LOAD
989 TO 1008 FX 1.2 FY -5.5
*3660-HOP-001
MEMBER LOAD
726 TO 728 730 CON GY -1
*3660-SCR-001
**FEEDBOX 8ton total
JOINT LOAD
876 877 880 884 FY -2
**CHT-001
875 876 FY -1
**SCR-001
882 886 889 893 FY -11
**CHT-002
883 887 888 892 FY -1.5
FLOOR LOAD
_NEWTOPFLOOR-1 FLOAD -0.0351 GY
_NEWTOPFLOOR-2 FLOAD -0.035 GY
_NEWPLATFORM FLOAD -0.035 GY
JOINT LOAD
1019 TO 1023 FY -2
FLOOR LOAD
_VALVEMAINTPLAT FLOAD -0.035 GY
****
****
LOAD 2 LOADTYPE Live TITLE LIVE-1 LOAD
**Grating Weight

FLOOR LOAD
_FLOOR-1 FLOAD -0.5 GY
_FLOOR-2_1 FLOAD -0.5 GY
_FLOOR-2_2 FLOAD -0.5 GY
_FLOOR-2_3 FLOAD -0.5 GY
_FLOOR-3_1 FLOAD -0.5 GY
_FLOOR-3_2 FLOAD -0.5 GY
_FLOOR-4_1 FLOAD -0.5 GY
_FLOOR-4_2 FLOAD -0.5 GY
_FLOOR-4_3 FLOAD -0.5 GY
_FLOOR-5_1 FLOAD -0.5 GY
_FLOOR-5_2 FLOAD -0.5 GY
_FLOOR-7_1 FLOAD -0.5 GY
_FLOOR-7_2 FLOAD -0.5 GY
_FLOOR-8_1 FLOAD -0.5 GY
_FLOOR-8_2 FLOAD -0.5 GY
_FLOOR-9_1 FLOAD -0.5 GY
_FLOOR-9_2 FLOAD -0.5 GY
MEMBER LOAD
15 20 140 158 167 216 CON GY -0.75 0.075
15 20 158 167 216 CON GY -0.75 1.275
1631 CON GY -0.75 0.5
58 654 CON GY -0.75 0.225
654 CON GY -0.75 1.425
276 CON GY -0.75 0.825
134 146 147 CON GY -0.75 0.35
134 146 147 CON GY -0.75 1.55
134 146 CON GY -0.75 1.85
134 146 CON GY -0.75 3.05
285 287 288 CON GY -0.75 0
287 CON GY -0.75 1.2
291 292 CON GY -0.75 0.125
*Monorail Weight
110 113 CON GY -5 1.8235
90 104 CON GY -5 1.636
132 143 CON GY -5 1.992
*Cable & Pipe Load
JOINT LOAD
33 34 41 42 54 55 58 59 64 65 75 76 79 80 935 936 FY -5
FLOOR LOAD
_NEWTOPFLOOR-1 FLOAD -0.5 GY
_NEWTOPFLOOR-2 FLOAD -0.5 GY
_NEWPLATFORM FLOAD -0.5 GY
_VALVEMAINTPLAT FLOAD -0.5 GY
****
LOAD 3 LOADTYPE None TITLE LIVE-2 LOAD
*3660-SCR-001
****
LOAD 4 LOADTYPE None TITLE LIVE-3 LOAD
MEMBER LOAD

*3660-TNK-003
399 400 403 404 CON GY -12
*3660-FAN-001 considered in DL
*3660-HEX-001 to 005
*3660-HOP-001
726 TO 728 730 CON GY -12.5
*3660-SCR-001
***
LOAD 5 LOADTYPE Wind TITLE WIND LOAD (+X DIR)
MEMBER LOAD
133 144 387 388 390 391 875 928 931 932 943 944 971 973 974 983 UNI GX 0.096
717 720 TO 722 724 TO 727 1060 UNI GX 0.085
132 143 395 398 TO 402 940 TO 942 961 962 986 UNI GX 0.077
3 4 6 10 11 21 43 TO 46 59 TO 62 77 TO 80 89 93 94 114 TO 116 135 TO 137 166 -
215 217 437 438 441 444 446 448 TO 450 511 514 898 912 1212 1214 TO 1216 -
1218 1219 1221 TO 1223 1225 1265 1267 1268 1270 1271 1273 1274 1276 1314 -
1352 1353 1367 1370 1372 1375 1663 TO 1665 UNI GX 0.065
119 134 139 140 146 147 152 TO 158 160 TO 165 167 216 260 270 271 281 TO 285 -
288 291 292 321 326 328 329 331 334 336 337 339 340 342 343 345 347 349 355 -
358 360 367 TO 376 379 380 383 TO 386 407 TO 410 423 425 430 432 539 550 -
554 560 562 569 571 572 575 580 585 586 588 590 594 595 597 599 603 605 607 -
608 TO 609 611 612 614 615 617 619 TO 621 623 625 626 631 636 637 643 645 -
646 648 650 665 667 669 TO 671 673 675 676 683 687 690 693 696 697 699 700 -
703 705 707 TO 709 711 712 714 732 TO 735 738 891 903 906 909 915 916 923 -
996 997 1012 1015 1028 1029 1042 1045 1057 TO 1059 1069 1070 1072 1172 1194 -
1195 1197 1200 1202 1204 1207 1209 1211 1227 1229 1231 1234 1236 1238 1260 -
1263 1278 1281 1284 1287 1308 1309 1312 1313 1315 TO 1318 1326 1328 1330 -
1336 1338 1340 1345 1346 1354 TO 1356 1368 1373 1409 1412 1416 1417 1425 -
1426 1537 1539 1631 1674 TO 1706 1708 1710 1712 1714 1716 1718 1721 1723 -
1744 1745 1766 TO 1775 UNI GX 0.052
1603 UNI GX 0.052 1.46167 1.87417
1603 UNI GX 0.052 1.87417 2.28667
1603 UNI GX 0.052 0.64167 1.46167
1603 UNI GX 0.052 0 0.64167
1421 UNI GX 0.052 0 0.17833
1600 UNI GX 0.052 0.13667 0.65667
1600 UNI GX 0.052 1.17667 1.47667
1410 UNI GX 0.052 0.4125 0.825
1410 UNI GX 0.052 0 0.4125
1600 UNI GX 0.052 0 0.13667
1410 UNI GX 0.052 0.825 0.98833
1600 UNI GX 0.052 0.65667 1.17667
1 2 8 9 23 24 26 27 41 42 53 TO 56 75 76 85 86 88 95 96 98 99 101 117 127 -
130 131 142 145 149 768 769 787 790 TO 792 885 1389 1390 1404 1405 1614 1615 -
1618 1621 1806 1807 UNI GX 0.032
876 877 929 930 UNI GX 0.028
102 105 106 108 109 111 112 851 852 UNI GX 0.026
118 845 846 848 TO 850 873 874 895 926 927 1169 1798 UNI GX 0.02
122 123 293 TO 296 847 853 TO 859 866 TO 869 871 872 878 TO 881 883 884 896 -
897 901 902 904 905 907 908 910 911 913 914 917 TO 922 924 925 1170 1171 -

1173 1174 1391 1394 1395 1397 1399 1402 1403 1616 1617 1619 1620 1623 1634 -
1635 1802 1803 UNI GX 0.016
141 148 159 168 218 899 900 1396 1398 1400 1401 1607 UNI GX 0.014
***Stair Weight
20 CON GX 0.25 0.075
15 20 CON GX 0.25 1.275
58 654 CON GX 0.25 0.225
654 CON GX 0.25 1.425
276 CON GX 0.25 0.825
287 CON GX 0.25 0
287 CON GX 0.25 1.2
*Monorail Weight
110 113 CON GX 0.6 1.8235
90 104 CON GX 0.6 1.636
*3660-TNK-003
399 400 403 404 CON GX 0.2
*3660-FAN-001
1216 1223 CON GX 0.2 0.319
1212 1219 CON GX 0.4 1.408
1270 1276 CON GX 0.2 0.4
1267 1273 CON GX 0.4 0.822
1367 1370 CON GX 0.2 0
*3660-HEX-001 to 005
1676 CON GX 0.01 0.32
1721 CON GX 0.01 0.14
1721 CON GX 0.01 0.5
1691 CON GX 0.01 0.0199999
1718 CON GX 0.01 0.29
1718 CON GX 0.01 0.65
432 CON GX 0.01 0.17
425 CON GX 0.01 0
425 CON GX 0.01 0.36
1681 CON GX 0.01 0.0700001
1710 CON GX 0.01 0.29
1710 CON GX 0.01 0.65
1695 CON GX 0.01 0.37
1706 CON GX 0.01 0.19
1706 CON GX 0.01 0.55
*3660-HOP-001
726 TO 728 730 CON GX 0.36
*3660-SCR-001
***
LOAD 6 LOADTYPE Wind TITLE WIND LOAD (-X DIR)
MEMBER LOAD
133 144 387 388 390 391 875 928 931 932 943 944 971 973 974 983 UNI GX -0.096
717 720 TO 722 724 TO 727 1060 UNI GX -0.085
132 143 395 398 TO 402 940 TO 942 961 962 986 UNI GX -0.077
3 4 6 10 11 21 43 TO 46 59 TO 62 77 TO 80 89 93 94 114 TO 116 135 TO 137 166 -
215 217 437 438 441 444 446 448 TO 450 511 514 898 912 1212 1214 TO 1216 -
1218 1219 1221 TO 1223 1225 1265 1267 1268 1270 1271 1273 1274 1276 1314 -

1352 1353 1367 1370 1372 1375 1663 TO 1665 UNI GX -0.065
119 134 139 140 146 147 152 TO 158 160 TO 165 167 216 260 270 271 281 TO 285 -
288 291 292 321 326 328 329 331 334 336 337 339 340 342 343 345 347 349 355 -
358 360 367 TO 376 379 380 383 TO 386 407 TO 410 423 425 430 432 539 550 -
554 560 562 569 571 572 575 580 585 586 588 590 594 595 597 599 603 605 607 -
608 TO 609 611 612 614 615 617 619 TO 621 623 625 626 631 636 637 643 645 -
646 648 650 665 667 669 TO 671 673 675 676 683 687 690 693 696 697 699 700 -
703 705 707 TO 709 711 712 714 732 TO 735 738 891 903 906 909 915 916 923 -
996 997 1012 1015 1028 1029 1042 1045 1057 TO 1059 1069 1070 1072 1172 1194 -
1195 1197 1200 1202 1204 1207 1209 1211 1227 1229 1231 1234 1236 1238 1260 -
1263 1278 1281 1284 1287 1308 1309 1312 1313 1315 TO 1318 1326 1328 1330 -
1336 1338 1340 1345 1346 1354 TO 1356 1368 1373 1409 1412 1416 1417 1425 -
1426 1537 1539 1631 1674 TO 1706 1708 1710 1712 1714 1716 1718 1721 1723 -
1744 1745 1766 TO 1775 UNI GX -0.052
1603 UNI GX -0.052 1.46167 1.87417
1603 UNI GX -0.052 1.87417 2.28667
1603 UNI GX -0.052 0.64167 1.46167
1603 UNI GX -0.052 0 0.64167
1421 UNI GX -0.052 0 0.17833
1600 UNI GX -0.052 0.13667 0.65667
1600 UNI GX -0.052 1.17667 1.47667
1410 UNI GX -0.052 0.4125 0.825
1410 UNI GX -0.052 0 0.4125
1600 UNI GX -0.052 0 0.13667
1410 UNI GX -0.052 0.825 0.98833
1600 UNI GX -0.052 0.65667 1.17667
1 2 8 9 23 24 26 27 41 42 53 TO 56 75 76 85 86 88 95 96 98 99 101 117 127 -
130 131 142 145 149 768 769 787 790 TO 792 885 1389 1390 1404 1405 1614 1615 -
1618 1621 1806 1807 UNI GX -0.032
876 877 929 930 UNI GX -0.028
102 105 106 108 109 111 112 851 852 UNI GX -0.026
118 845 846 848 TO 850 873 874 895 926 927 1169 1798 UNI GX -0.02
122 123 293 TO 296 847 853 TO 859 866 TO 869 871 872 878 TO 881 883 884 896 -
897 901 902 904 905 907 908 910 911 913 914 917 TO 922 924 925 1170 1171 -
1173 1174 1391 1394 1395 1397 1399 1402 1403 1616 1617 1619 1620 1623 1634 -
1635 1802 1803 UNI GX -0.016
141 148 159 168 218 899 900 1396 1398 1400 1401 1607 UNI GX -0.014
***Stair Weight
20 CON GX -0.25 0.075
15 20 CON GX -0.25 1.275
58 654 CON GX -0.25 0.225
654 CON GX -0.25 1.425
276 CON GX -0.25 0.825
287 CON GX -0.25 0
287 CON GX -0.25 1.2
*Monorail Weight
110 113 CON GX -0.6 1.8235
90 104 CON GX -0.6 1.636
*3660-TNK-003
399 400 403 404 CON GX -0.2

*3660-FAN-001
1216 1223 CON GX -0.2 0.319
1212 1219 CON GX -0.4 1.408
1270 1276 CON GX -0.2 0.4
1267 1273 CON GX -0.4 0.822
1367 1370 CON GX -0.2 0
*3660-HEX-001 to 005
1676 CON GX -0.01 0.32
1721 CON GX -0.01 0.14
1721 CON GX -0.01 0.5
1691 CON GX -0.01 0.0199999
1718 CON GX -0.01 0.29
1718 CON GX -0.01 0.65
432 CON GX -0.01 0.17
425 CON GX -0.01 0
425 CON GX -0.01 0.36
1681 CON GX -0.01 0.0700001
1710 CON GX -0.01 0.29
1710 CON GX -0.01 0.65
1695 CON GX -0.01 0.37
1706 CON GX -0.01 0.19
1706 CON GX -0.01 0.55
*3660-HOP-001
726 TO 728 730 CON GX -0.36
*3660-SCR-001
***
LOAD 7 LOADTYPE Wind TITLE WIND LOAD(+Z DIR)
MEMBER LOAD
389 392 TO 394 396 397 403 TO 406 958 959 UNI GZ 0.096
28 51 715 716 718 719 728 729 805 1063 1064 1179 1406 1408 1413 TO 1415 1479 -
1480 1485 TO 1488 1515 TO 1518 1599 UNI GZ 0.085
87 90 100 104 110 113 723 730 731 820 842 936 TO 938 951 952 957 -
960 UNI GZ 0.077
1 2 8 9 12 13 23 24 26 27 41 42 48 49 53 TO 56 75 76 81 82 85 86 88 -
95 TO 99 101 117 127 130 131 142 145 149 421 422 426 428 429 433 435 436 -
439 440 442 443 445 447 538 768 769 777 787 790 TO 792 797 823 885 1196 1198 -
1199 1201 1203 1205 1206 1208 1210 1305 TO 1307 1310 1311 1386 1389 1390 -
1404 1405 1614 1615 1618 1621 1806 1807 UNI GZ 0.065
7 14 15 19 20 39 47 50 57 58 63 TO 66 83 91 92 102 105 106 108 109 111 112 -
150 276 286 287 316 319 323 325 327 330 332 335 338 341 344 346 348 351 353 -
356 359 362 TO 366 371 TO 374 377 378 381 TO 386 407 TO 410 414 417 552 555 -
556 561 564 570 573 574 577 579 581 587 591 596 598 600 604 TO 606 608 610 -
611 613 614 616 TO 618 620 622 TO 624 626 632 639 641 644 647 651 654 656 -
658 666 TO 668 670 672 TO 674 676 684 685 688 689 692 695 698 701 702 704 -
705 TO 706 708 710 711 713 714 732 TO 735 851 852 985 990 991 993 994 1007 -
1009 1010 1022 1023 1025 1026 1037 1039 1040 1052 1055 1175 1357 TO 1362 -
1369 1371 1374 1376 1637 TO 1644 1649 TO 1652 1657 1659 1661 1662 1739 1741 -
1743 UNI GZ 0.052
3 4 6 10 11 21 43 TO 46 59 TO 62 77 TO 80 89 93 94 166 215 217 898 912 1663 -
1664 TO 1665 UNI GZ 0.032

281 TO 284 UNI GZ 0.026
25 34 778 779 788 793 TO 796 798 799 801 811 819 862 TO 865 1176 TO 1178 1180 -
1181 1377 1378 1385 1387 1388 1605 1606 1610 TO 1613 UNI GZ 0.02
67 TO 74 289 290 780 783 TO 786 821 822 824 825 828 832 TO 841 843 844 1381 -
1383 UNI GZ 0.016
16 22 40 151 812 814 TO 816 1054 1379 1380 1382 1384 UNI GZ 0.014
***Stair Weight
140 158 167 216 CON GZ 0.25 0.075
158 167 216 CON GZ 0.25 1.275
1631 CON GZ 0.25 0.5
134 146 147 CON GZ 0.25 0.35
134 146 147 CON GZ 0.25 1.55
134 146 CON GZ 0.25 1.85
134 146 CON GZ 0.25 3.05
285 288 CON GZ 0.25 0
291 292 CON GZ 0.25 0.125
*Monorail Weight
132 143 CON GZ 0.6 1.992
*3660-TNK-003
399 400 403 404 CON GZ 0.2
*3660-FAN-001
1216 1223 CON GZ 0.07 0.319
1212 1219 CON GZ 0.14 1.408
1270 1276 CON GZ 0.07 0.4
1267 1273 CON GZ 0.14 0.822
1367 1370 CON GZ 0.07 0
*3660-HEX-001 to 005
1676 CON GZ 0.01 0.32
1721 CON GZ 0.01 0.14
1721 CON GZ 0.01 0.5
1691 CON GZ 0.01 0.0199999
1718 CON GZ 0.01 0.29
1718 CON GZ 0.01 0.65
432 CON GZ 0.01 0.17
425 CON GZ 0.01 0
425 CON GZ 0.01 0.36
1681 CON GZ 0.01 0.0700001
1710 CON GZ 0.01 0.29
1710 CON GZ 0.01 0.65
1695 CON GZ 0.01 0.37
1706 CON GZ 0.01 0.19
1706 CON GZ 0.01 0.55
*3660-HOP-001
726 TO 728 730 CON GZ 0.36
*3660-SCR-001
***
LOAD 8 LOADTYPE Wind TITLE WIND LOAD(-Z DIR)
MEMBER LOAD
389 392 TO 394 396 397 403 TO 406 958 959 UNI GZ -0.096
28 51 715 716 718 719 728 729 805 1063 1064 1179 1406 1408 1413 TO 1415 1479 -

1480 1485 TO 1488 1515 TO 1518 1599 UNI GZ -0.085
87 90 100 104 110 113 723 730 731 820 842 936 TO 938 951 952 957 -
960 UNI GZ -0.077
1 2 8 9 12 13 23 24 26 27 41 42 48 49 53 TO 56 75 76 81 82 85 86 88 -
95 TO 99 101 117 127 130 131 142 145 149 421 422 426 428 429 433 435 436 -
439 440 442 443 445 447 538 768 769 777 787 790 TO 792 797 823 885 1196 1198 -
1199 1201 1203 1205 1206 1208 1210 1305 TO 1307 1310 1311 1386 1389 1390 -
1404 1405 1614 1615 1618 1621 1806 1807 UNI GZ -0.065
7 14 15 19 20 39 47 50 57 58 63 TO 66 83 91 92 102 105 106 108 109 111 112 -
150 276 286 287 316 319 323 325 327 330 332 335 338 341 344 346 348 351 353 -
356 359 362 TO 366 371 TO 374 377 378 381 TO 386 407 TO 410 414 417 552 555 -
556 561 564 570 573 574 577 579 581 587 591 596 598 600 604 TO 606 608 610 -
611 613 614 616 TO 618 620 622 TO 624 626 632 639 641 644 647 651 654 656 -
658 666 TO 668 670 672 TO 674 676 684 685 688 689 692 695 698 701 702 704 -
705 TO 706 708 710 711 713 714 732 TO 735 851 852 985 990 991 993 994 1007 -
1009 1010 1022 1023 1025 1026 1037 1039 1040 1052 1055 1175 1357 TO 1362 -
1369 1371 1374 1376 1637 TO 1644 1649 TO 1652 1657 1659 1661 1662 1739 1741 -
1743 UNI GZ -0.052
3 4 6 10 11 21 43 TO 46 59 TO 62 77 TO 80 89 93 94 166 215 217 898 912 1663 -
1664 TO 1665 UNI GZ -0.032
281 TO 284 UNI GZ -0.026
25 34 778 779 788 793 TO 796 798 799 801 811 819 862 TO 865 1176 TO 1178 1180 -
1181 1377 1378 1385 1387 1388 1605 1606 1610 TO 1613 UNI GZ -0.02
67 TO 74 289 290 780 783 TO 786 821 822 824 825 828 832 TO 841 843 844 1381 -
1383 UNI GZ -0.016
16 22 40 151 812 814 TO 816 1054 1379 1380 1382 1384 UNI GZ -0.014
***Stair Weight
140 158 167 216 CON GZ -0.25 0.075
158 167 216 CON GZ -0.25 1.275
1631 CON GZ -0.25 0.5
134 146 147 CON GZ -0.25 0.35
134 146 147 CON GZ -0.25 1.55
134 146 CON GZ -0.25 1.85
134 146 CON GZ -0.25 3.05
285 288 CON GZ -0.25 0
291 292 CON GZ -0.25 0.125
*Monorail Weight
132 143 CON GZ -0.6 1.992
*3660-TNK-003
399 400 403 404 CON GZ -0.2
*3660-FAN-001
1216 1223 CON GZ -0.07 0.319
1212 1219 CON GZ -0.14 1.408
1270 1276 CON GZ -0.07 0.4
1267 1273 CON GZ -0.14 0.822
1367 1370 CON GZ -0.07 0
*3660-HEX-001 to 005
1676 CON GZ -0.01 0.32
1721 CON GZ -0.01 0.14
1721 CON GZ -0.01 0.5

```
1691 CON GZ -0.01 0.0199999
1718 CON GZ -0.01 0.29
1718 CON GZ -0.01 0.65
432 CON GZ -0.01 0.17
425 CON GZ -0.01 0
425 CON GZ -0.01 0.36
1681 CON GZ -0.01 0.0700001
1710 CON GZ -0.01 0.29
1710 CON GZ -0.01 0.65
1695 CON GZ -0.01 0.37
1706 CON GZ -0.01 0.19
1706 CON GZ -0.01 0.55
*3660-HOP-001
726 TO 728 730 CON GZ -0.36
*3660-SCR-001
***
************************************************************
****************************
*****************************************************************************
*
**********LOAD COMBS FOR
DESIGN*******************************************
LOAD 11 LOADTYPE Dead  TITLE DL
REPEAT LOAD
1 1.2
LOAD 12 LOADTYPE Live  TITLE LL1
REPEAT LOAD
2 1.2
LOAD 13 LOADTYPE Live  TITLE LL2
REPEAT LOAD
3 1.2
LOAD 14 LOADTYPE Live  TITLE LL3
REPEAT LOAD
4 1.2
LOAD 15 LOADTYPE Wind  TITLE WL1
REPEAT LOAD
5 1.2
LOAD 16 LOADTYPE Wind  TITLE WL2
REPEAT LOAD
6 1.2
LOAD 17 LOADTYPE Wind  TITLE WL3
REPEAT LOAD
7 1.2
LOAD 18 LOADTYPE Wind  TITLE WL4
REPEAT LOAD
8 1.2
*SERVICE LOAD COMBINATIONS*
LOAD COMB 101 1.4 DL
11 1.4
LOAD COMB 102 1.2 (DL) + 1.6 (LL1+LL2+LL3)
```

11 1.2 12 1.6 13 1.6 14 1.6
LOAD COMB 103 1.2 DL+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
11 1.2 15 1.6 12 0.5 13 0.5 14 0.5
LOAD COMB 104 1.2 DL+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)
11 1.2 16 1.6 12 0.5 13 0.5 14 0.5
LOAD COMB 105 1.2 DL+ 1.6 WL3 + 0.5 (LL1+LL2+LL3)
11 1.2 17 1.6 12 0.5 13 0.5 14 0.5
LOAD COMB 106 1.2 DL+ 1.6 WL4 + 0.5 (LL1+LL2+LL3)
11 1.2 18 1.6 12 0.5 13 0.5 14 0.5
LOAD COMB 107 0.9 DL + 1.6 WL1
11 0.9 15 1.6
LOAD COMB 108 0.9 DL + 1.6 WL2
11 0.9 16 1.6
LOAD COMB 109 0.9 DL + 1.6 WL3
11 0.9 17 1.6
LOAD COMB 110 0.9 DL + 1.6 WL4
11 0.9 18 1.6
LOAD COMB 111 1.2 DL + 0.8 WL1
11 1.2 15 0.8
LOAD COMB 112 1.2 DL + 0.8 WL2
11 1.2 16 0.8
LOAD COMB 113 1.2 DL + 0.8 WL3
11 1.2 17 0.8
LOAD COMB 114 1.2 DL + 0.8 WL4
11 1.2 18 0.8
****
****Serviceanbility check
LOAD COMB 201 1. DL
11 1.0
LOAD COMB 202 1 DL + 1 (LL1+LL2+LL3)
11 1.0 12 1.0 13 1.0 14 1.0
LOAD COMB 203 1 DL+ 1 WL1
11 1.0 15 1.0
LOAD COMB 204 1 DL+ 1 WL2
11 1.0 16 1.0
LOAD COMB 205 1 DL+ 1 WL3
11 1.0 17 1.0
LOAD COMB 206 1 DL+ 1 WL4
11 1.0 18 1.0
LOAD COMB 207 0.6 DL + 1 WL1
11 0.6 15 1.0
LOAD COMB 208 0.6 DL + 1 WL2
11 0.6 16 1.0
LOAD COMB 209 0.6 DL + 1 WL3
11 0.6 17 1.0
LOAD COMB 210 0.6 DL + 1 WL4
11 0.6 18 1.0
LOAD COMB 211 1 DL+ 0.75 WL1+ 0.75 (LL1+LL2+LL3)
11 1.0 15 0.75 12 0.75 13 0.75 14 0.75
LOAD COMB 212 1 DL+ 0.75 WL2+ 0.75 (LL1+LL2+LL3)

11 1.0 16 0.75 12 0.75 13 0.75 14 0.75
LOAD COMB 213 1 DL+ 0.75 WL3+ 0.75 (LL1+LL2+LL3)
11 1.0 17 0.75 12 0.75 13 0.75 14 0.75
LOAD COMB 214 1 DL+ 0.75 WL4+ 0.75 (LL1+LL2+LL3)
11 1.0 18 0.75 12 0.75 13 0.75 14 0.75
*3660-SCR-001
***
****
****
**FOUNDATION LOAD DATA
****
*
PERFORM ANALYSIS
FINISH

ATTACHMENT 2 STAAD INPUT FOR FOOTING F3:

PROPRIETARY & CONFIDENTIAL

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-May-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 2.3 0 0; 3 2.3 0 2.3; 4 0 0 2.3; 5 0 2.105 0; 6 2.3 2.105 0;
7 2.3 2.105 2.3; 8 0 2.105 2.3; 9 0 2.105 1.149; 10 1.149 2.105 0;
11 2.3 2.105 1.149; 12 1.149 2.105 2.3; 13 1.149 9.105 1.149; 14 0 -2.05 0;
15 2.3 -2.05 0; 16 2.3 -2.05 2.3; 17 0 -2.05 2.3; 18 -1.55 -2.05 -1.05;
19 3.45 -2.05 -1.05; 20 -1.55 -2.05 2.95; 21 3.45 -2.05 2.95;
22 -1.55 -2.05 7.45; 23 3.45 -2.05 7.45; 24 0 0 4.15; 25 0 2.105 4.15;
26 0 0 6.45; 27 0 2.105 6.45; 28 2.3 0 4.15; 29 2.3 2.105 4.15; 30 2.3 0 6.45;
31 2.3 2.105 6.45; 32 0 2.105 5.299; 33 1.149 2.105 6.45; 34 2.3 2.105 5.299;
35 1.149 2.105 4.15; 36 1.149 9.105 5.299; 37 0 -2.05 4.15; 38 0 -2.05 6.45;
39 2.3 -2.05 4.15; 40 2.3 -2.05 6.45; 41 -1.55 -0.25 -1.05;
42 3.45 -0.25 -1.05; 43 -1.55 -0.25 2.95; 44 3.45 -0.25 2.95;
45 -1.55 -0.25 7.45; 46 3.45 -0.25 7.45; 47 -2.05 -2.05 -1.55;
48 3.95 -2.05 -1.55; 49 -2.05 -2.05 7.95; 50 3.95 -2.05 7.95;
51 -1.95 -2.05 -1.55; 52 -1.95 -2.05 -1.45; 53 -2.05 -2.05 -1.45;
54 -1.85 -2.05 -1.55; 55 -1.85 -2.05 -1.45; 56 -1.75 -2.05 -1.55;
57 -1.75 -2.05 -1.45; 58 -1.65 -2.05 -1.55; 59 -1.65 -2.05 -1.45;
60 -1.55 -2.05 -1.55; 61 -1.55 -2.05 -1.45; 62 -1.45 -2.05 -1.55;
63 -1.45 -2.05 -1.45; 64 -1.35 -2.05 -1.55; 65 -1.35 -2.05 -1.45;
66 -1.25 -2.05 -1.55; 67 -1.25 -2.05 -1.45; 68 -1.15 -2.05 -1.55;
69 -1.15 -2.05 -1.45; 70 -1.05 -2.05 -1.55; 71 -1.05 -2.05 -1.45;
72 -0.95 -2.05 -1.55; 73 -0.95 -2.05 -1.45; 74 -0.85 -2.05 -1.55;
75 -0.85 -2.05 -1.45; 76 -0.75 -2.05 -1.55; 77 -0.75 -2.05 -1.45;
78 -0.65 -2.05 -1.55; 79 -0.65 -2.05 -1.45; 80 -0.55 -2.05 -1.55;
81 -0.55 -2.05 -1.45; 82 -0.45 -2.05 -1.55; 83 -0.45 -2.05 -1.45;
84 -0.35 -2.05 -1.55; 85 -0.35 -2.05 -1.45; 86 -0.25 -2.05 -1.55;
87 -0.25 -2.05 -1.45; 88 -0.15 -2.05 -1.55; 89 -0.15 -2.05 -1.45;
90 -0.0499999 -2.05 -1.55; 91 -0.0499999 -2.05 -1.45; 92 0.0500002 -2.05 -1.55;
93 0.0500002 -2.05 -1.45; 94 0.15 -2.05 -1.55; 95 0.15 -2.05 -1.45;
96 0.25 -2.05 -1.55; 97 0.25 -2.05 -1.45; 98 0.35 -2.05 -1.55;
99 0.35 -2.05 -1.45; 100 0.45 -2.05 -1.55; 101 0.45 -2.05 -1.45;
102 0.55 -2.05 -1.55; 103 0.55 -2.05 -1.45; 104 0.65 -2.05 -1.55;
105 0.65 -2.05 -1.45; 106 0.75 -2.05 -1.55; 107 0.75 -2.05 -1.45;
108 0.85 -2.05 -1.55; 109 0.85 -2.05 -1.45; 110 0.95 -2.05 -1.55;
111 0.95 -2.05 -1.45; 112 1.05 -2.05 -1.55; 113 1.05 -2.05 -1.45;
114 1.15 -2.05 -1.55; 115 1.15 -2.05 -1.45; 116 1.25 -2.05 -1.55;
117 1.25 -2.05 -1.45; 118 1.35 -2.05 -1.55; 119 1.35 -2.05 -1.45;
120 1.45 -2.05 -1.55; 121 1.45 -2.05 -1.45; 122 1.55 -2.05 -1.55;
123 1.55 -2.05 -1.45; 124 1.65 -2.05 -1.55; 125 1.65 -2.05 -1.45;
126 1.75 -2.05 -1.55; 127 1.75 -2.05 -1.45; 128 1.85 -2.05 -1.55;
129 1.85 -2.05 -1.45; 130 1.95 -2.05 -1.55; 131 1.95 -2.05 -1.45;
132 2.05 -2.05 -1.55; 133 2.05 -2.05 -1.45; 134 2.15 -2.05 -1.55;
135 2.15 -2.05 -1.45; 136 2.25 -2.05 -1.55; 137 2.25 -2.05 -1.45;
138 2.35 -2.05 -1.55; 139 2.35 -2.05 -1.45; 140 2.45 -2.05 -1.55;

141 2.45 -2.05 -1.45; 142 2.55 -2.05 -1.55; 143 2.55 -2.05 -1.45;
144 2.65 -2.05 -1.55; 145 2.65 -2.05 -1.45; 146 2.75 -2.05 -1.55;
147 2.75 -2.05 -1.45; 148 2.85 -2.05 -1.55; 149 2.85 -2.05 -1.45;
150 2.95 -2.05 -1.55; 151 2.95 -2.05 -1.45; 152 3.05 -2.05 -1.55;
153 3.05 -2.05 -1.45; 154 3.15 -2.05 -1.55; 155 3.15 -2.05 -1.45;
156 3.25 -2.05 -1.55; 157 3.25 -2.05 -1.45; 158 3.35 -2.05 -1.55;
159 3.35 -2.05 -1.45; 160 3.45 -2.05 -1.55; 161 3.45 -2.05 -1.45;
162 3.55 -2.05 -1.55; 163 3.55 -2.05 -1.45; 164 3.65 -2.05 -1.55;
165 3.65 -2.05 -1.45; 166 3.75 -2.05 -1.55; 167 3.75 -2.05 -1.45;
168 3.85 -2.05 -1.55; 169 3.85 -2.05 -1.45; 170 3.95 -2.05 -1.45;
171 -1.95 -2.05 -1.35; 172 -2.05 -2.05 -1.35; 173 -1.85 -2.05 -1.35;
174 -1.75 -2.05 -1.35; 175 -1.65 -2.05 -1.35; 176 -1.55 -2.05 -1.35;
177 -1.45 -2.05 -1.35; 178 -1.35 -2.05 -1.35; 179 -1.25 -2.05 -1.35;
180 -1.15 -2.05 -1.35; 181 -1.05 -2.05 -1.35; 182 -0.95 -2.05 -1.35;
183 -0.85 -2.05 -1.35; 184 -0.75 -2.05 -1.35; 185 -0.65 -2.05 -1.35;
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MEMBER INCIDENCES
1 1 5; 2 4 8; 3 2 6; 4 3 7; 5 5 9; 6 8 12; 7 6 11; 8 5 10; 9 9 8; 10 10 6;
11 11 7; 12 12 7; 13 1 9; 14 4 9; 15 4 12; 16 3 12; 17 1 10; 18 2 10; 19 2 11;
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ELEMENT PROPERTY

64 TO 5763 THICKNESS 0.7
UNIT METER MTON
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40261
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
G 946438
TYPE CONCRETE
STRENGTH FCU 2855.21
STRENGTH FCU 2812.28
ISOTROPIC RIGID_STEEL
E 1.01972e+009
POISSON 0.17
DENSITY 7.85181
ALPHA 1.2e-005
DAMP 0.03
G 7.95379e+006
END DEFINE MATERIAL
UNIT METER KN
MEMBER PROPERTY BRITISH
5 TO 12 33 TO 40 TABLE ST UB203X102X23
1 TO 4 29 TO 32 TABLE ST UC203X203X46
MEMBER PROPERTY EUROPEAN
13 TO 20 41 TO 48 TABLE ST 101.6X4CHS
MEMBER PROPERTY EUROPEAN
21 TO 24 49 TO 52 5764 TO 5787 PRIS YD 0.5
MEMBER PROPERTY EUROPEAN
25 TO 28 53 TO 56 PRIS YD 0.6 ZD 0.6
MEMBER PROPERTY EUROPEAN
57 TO 62 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL STEEL MEMB 1 TO 20 29 TO 48
MATERIAL RIGID_STEEL MEMB 21 TO 24 49 TO 52 5764 TO 5787
MATERIAL CONCRETE MEMB 25 TO 28 53 TO 62 64 TO 5763
MEMBER RELEASE
5 TO 8 13 TO 24 33 TO 36 41 TO 52 START MY MZ
9 TO 24 37 TO 52 END MY MZ
*DEFINE IBC 2003

*SDS 0.06 SD1 0.013 S1 0.02 IE 1 RX 3 RZ 4 SCLASS 2 PX 0.332 PZ 0.332
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SUPPORTS
64 TO 5763 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
LOAD 1 LOADTYPE Dead TITLE DL1_PUMP_OPERATING
JOINT LOAD
13 36 FY -541
LOAD 2 LOADTYPE Dead TITLE DL2_PUMP_EMPTY
JOINT LOAD
13 36 FY -513
LOAD 3 LOADTYPE Dead TITLE DL3_STRUCTURE
JOINT LOAD
41 FX 13.996 FY -103.576 FZ -0.598
43 FX 64.61 FY -343.133 FZ -1.529
45 FX 18.294 FY -183.933 FZ 6.253
42 FX 17.417 FY -61.358 FZ -0.051
44 FX 69.266 FY -141.154 FZ -2.652
46 FX 18.193 FY -72.196 FZ 1.605
LOAD 4 LOADTYPE Wind TITLE WL1
JOINT LOAD
13 36 FX 30
41 FX 24.308 FY 60.956
43 FX 57.725 FY 78.646 FZ 1.59
45 FX 27.062 FY 55.048 FZ 1.366
42 FX 24.371 FY -60.593 FZ -0.014
44 FX 57.793 FY -111.733 FZ -1.433
46 FX 27.158 FY -55.384 FZ -1.089
LOAD 5 LOADTYPE Wind TITLE WL2
JOINT LOAD
13 36 FX -30
41 FX -24.308 FY -60.956
43 FX -57.725 FY -78.646 FZ -1.59
45 FX -27.062 FY -55.048 FZ -1.366
42 FX -24.371 FY 60.593 FZ 0.014
44 FX -57.793 FY 111.733 FZ 1.433
46 FX -27.158 FY 55.384 FZ 1.089
LOAD 6 LOADTYPE Wind TITLE WL3
JOINT LOAD
13 36 FZ 30
41 FX 1.509 FY 73.223 FZ 0.964
43 FX -1.163 FY -17.011 FZ 39.583
45 FX -0.969 FY -48.259 FZ 39.362
42 FX 1.044 FY 67.308 FZ 1.13
44 FX -0.701 FY -24.794 FZ 39.924
46 FX -0.97 FY -45.019 FZ 39.751
LOAD 7 LOADTYPE Wind TITLE WL4
JOINT LOAD

13 36 FZ -30
41 FX -1.509 FY -73.223 FZ -0.964
43 FX 1.163 FY 17.011 FZ -39.583
45 FX 0.969 FY 48.259 FZ -39.362
42 FX -1.044 FY -67.305 FZ 1.13
44 FX 0.701 FY 24.794 FZ -39.924
LOAD 8 LOADTYPE Live TITLE LL1
JOINT LOAD
41 FX -3.255 FY -163.532 FZ -0.432
43 FX -7.411 FY -330.142 FZ -13.953
45 FX 0.091 FY -264.293 FZ 11.494
42 FX 7.115 FY -126.471 FZ -0.294
44 FX 10.588 FY -285.954 FZ -9.795
46 FX 0.625 FY -165.043 FZ 9.41
LOAD 9 LOADTYPE Live TITLE LL2
LOAD 10 LOADTYPE Live TITLE LL3
JOINT LOAD
41 FX -0.098 FY -0.76 FZ -0.015
43 FX 1.148 FY -5.393 FZ -0.295
45 FX -0.352 FY -0.46 FZ -0.26
42 FX -0.093 FY 0.376 FZ 0.004
44 FX 1.156 FY 0.823 FZ 0.028
46 FX -0.353 FY 0.759 FZ 0.018
LOAD 11 LOADTYPE Seismic TITLE EL1
JOINT LOAD
13 36 FX 54.1
LOAD 12 LOADTYPE Seismic TITLE EL2
JOINT LOAD
13 36 FZ 54.1
LOAD 13 LOADTYPE Seismic TITLE SL1_EMP(+X)
JOINT LOAD
13 36 FX 8
LOAD 14 LOADTYPE Seismic TITLE SL2_EMP(+Z)
JOINT LOAD
13 36 FZ 8
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LOAD COMB 101 1.4 (DL1+DL3)
1 1.4 3 1.4
LOAD COMB 102 1.2 (DL1+DL3) + 1.6 (LL1+LL2+LL3)
1 1.2 3 1.2 8 1.6 9 1.6 10 1.6
LOAD COMB 103 1.2 (DL1+DL3)+ 1.6 WL1 + 0.5 (LL1+LL2+LL3)
1 1.2 3 1.2 4 1.6 0.5
LOAD COMB 104 1.2 (DL1+DL3)+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)
1 1.2 3 1.2 5 1.6 0.5
LOAD COMB 105 1.2 (DL1+DL3)+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)
1 1.2 3 1.2 6 1.6 0.5
LOAD COMB 106 1.2 (DL1+DL3)+ 1.6 WL2 + 0.5 (LL1+LL2+LL3)
1 1.2 3 1.2 7 1.6 0.5
LOAD COMB 107 1.2 (DL1+DL3) + 1.0 EL1
1 1.2 3 1.2 11 1.0

LOAD COMB 108 1.2 (DL1+DL3) - 1.0 EL1
1 1.2 3 1.2 11 -1.0
LOAD COMB 109 1.2 (DL1+DL3) + 1.0 EL2
1 1.2 3 1.2 12 1.0
LOAD COMB 110 1.2 (DL1+DL3) - 1.0 EL2
1 1.2 3 1.2 12 -1.0
LOAD COMB 111 0.9 (DL2+DL3) + 1.6 WL1
3 0.9 2 0.9 4 1.6
LOAD COMB 112 0.9 (DL2+DL3) + 1.6 WL2
3 0.9 2 0.9 5 1.6
LOAD COMB 113 0.9 (DL2+DL3) + 1.6 WL3
3 0.9 2 0.9 6 1.6
LOAD COMB 114 0.9 (DL2+DL3) + 1.6 WL4
3 0.9 2 0.9 7 1.6
LOAD COMB 115 0.9 (DL2+DL3) + 1.0 SL1_EMP(+X)
2 0.9 3 0.9 13 1.0
LOAD COMB 116 0.9 (DL2+DL3) - 1.0 SL1_EMP(+X)
2 0.9 3 0.9 13 -1.0
LOAD COMB 117 0.9 (DL2+DL3) + 1.0 SL1_EMP(+Z)
2 0.9 3 0.9 14 1.0
LOAD COMB 118 0.9 (DL2+DL3) - 1.0 SL1_EMP(+Z)
2 0.9 3 0.9 14 -1.0
LOAD COMB 119 1.2 (DL1+DL3) + 0.8 WL1
1 1.2 3 1.2 4 0.8
LOAD COMB 120 1.2 (DL1+DL3) + 0.8 WL2
1 1.2 3 1.2 5 0.8
LOAD COMB 121 1.2 (DL1+DL3) + 0.8 WL3
1 1.2 3 1.2 6 0.8
LOAD COMB 122 1.2 (DL1+DL3) + 0.8 WL4
1 1.2 3 1.2 7 0.8
****
****Serviceanbility check
LOAD COMB 201 1. (DL1+DL3)
1 1.0 3 1.0
LOAD COMB 202 1 (DL1 + DL3) + 1 (LL1+LL2+LL3)
1 1.0 3 1.0 8 1.0 9 1.0 10 1.0
LOAD COMB 203 1 (DL+ DL3) + 1 WL1
1 1.0 3 1.0 4 1.0
LOAD COMB 204 1 (DL+ DL3) + 1 WL2
1 1.0 3 1.0 5 1.0
LOAD COMB 205 1 (DL+ DL3) + 1 WL3
1 1.0 3 1.0 6 1.0
LOAD COMB 206 1 (DL+ DL3) + 1 WL4
1 1.0 3 1.0 7 1.0
LOAD COMB 207 1 (DL1+DL3) + 0.7 EL1
1 1.0 3 1.0 11 0.7
LOAD COMB 208 1 (DL1+DL3) - 0.7 EL1
1 1.0 3 1.0 11 -0.7
LOAD COMB 209 1 (DL1+DL3) + 0.7 EL2
1 1.0 3 1.0 12 0.7

LOAD COMB 210 1 (DL1+DL3) - 0.7 EL2
1 1.0 3 1.0 12 -0.7
LOAD COMB 211 0.6 (DL2+DL3) + 1 WL1
2 1.0 3 0.6 4 1.0
LOAD COMB 212 0.6 (DL2+DL3) + 1 WL2
2 1.0 3 0.6 5 1.0
LOAD COMB 213 0.6 (DL2+DL3) + 1 WL3
2 1.0 3 0.6 6 1.0
LOAD COMB 214 0.6 (DL2+DL3) + 1 WL4
2 1.0 3 0.6 7 1.0
LOAD COMB 215 0.6 (DL2+DL3) + 0.7 SL1_EMP(+X)
2 1.0 3 0.6 13 0.7
LOAD COMB 216 0.6 (DL2+DL3) - 0.7 SL1_EMP(+X)
2 1.0 3 0.6 13 -0.7
LOAD COMB 217 0.6 (DL2+DL3) + 0.7 SL1_EMP(+Z)
2 1.0 3 0.6 14 0.7
LOAD COMB 218 0.6 (DL2+DL3) - 0.7 SL1_EMP(+Z)
2 1.0 3 0.6 14 -0.7
LOAD COMB 219 1 (DL1+DL3)+ 0.75 WL1 + 0.75(LL1+LL2+LL3)
1 1.0 3 1.0 4 0.75 8 0.75 9 0.75 10 0.75
LOAD COMB 220 1 (DL1+DL3)+ 0.75 WL2 + 0.75(LL1+LL2+LL3)
1 1.0 3 1.0 5 0.75 8 0.75 9 0.75 10 0.75
LOAD COMB 221 1 (DL1+DL3)+ 0.75 WL3 + 0.75(LL1+LL2+LL3)
1 1.0 3 1.0 6 0.75 8 0.75 9 0.75 10 0.75
LOAD COMB 222 1 (DL1+DL3)+ 0.75 WL4 + 0.75(LL1+LL2+LL3)
1 1.0 3 1.0 7 0.75 8 0.75 9 0.75 10 0.75
LOAD COMB 223 1 (DL1 + DL3) + 0.525 EL1
1 1.0 3 1.0 11 0.525
LOAD COMB 224 1 (DL1 + DL3) - 0.525 EL1
1 1.0 3 1.0 11 -0.525
LOAD COMB 225 1 (DL1 + DL3) + 0.525 EL2
1 1.0 3 1.0 12 0.525
LOAD COMB 226 1 (DL1 + DL3) - 0.525 EL2
1 1.0 3 1.0 12 -0.525
****
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
FC 28000 ALL
FYMAIN 414000 ALL
MINMAIN 16 ALL
MINSEC 10 ALL
DESIGN COLUMN ALL
END CONCRETE DESIGN
***
FINISH

ATTACHMENT 3 STAAD INPUT FOR FOOTING F9 DESIGN:

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-May-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
494 -6.66 -1.67 -4.145; 495 -6.66 -2.87 -4.145; 496 -6.66 -0.27 -4.145;
497 -6.66 -1.67 2.605; 498 -6.66 -2.87 2.605; 499 -6.66 -0.27 2.605;
500 -6.66 -1.67 7.105; 501 -6.66 -2.87 7.105; 502 -6.66 -0.27 7.105;
503 -6.66 -1.67 11.105; 504 -6.66 -2.87 11.105; 505 -6.66 -0.27 11.105;
MEMBER INCIDENCES
551 494 495; 552 494 496; 553 497 498; 554 497 499; 555 500 501; 556 500 502;
557 503 504; 558 503 505;
*START GROUP DEFINITION
*FLOOR
*END GROUP DEFINITION
SUPPORTS
495 498 501 504 FIXED
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.21467e+006
POISSON 0.17
DENSITY 2.40261
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
G 946438
TYPE CONCRETE
STRENGTH FCU 2855.21
STRENGTH FCU 2812.28
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
551 TO 558 PRIS YD 0.7 ZD 0.7
CONSTANTS
MATERIAL CONCRETE ALL
*****
LOAD 1 LOADTYPE Dead  TITLE DL
UNIT METER KN
SELFWEIGHT Y -1
JOINT LOAD
505 FX 21.425 FY -318.05 FZ -3.248
```

502 FX -12.325 FY -173.6 FZ -5.1
499 FX 5.973 FY -73.103 FZ 25.103
496 FX -2.432 FY -70.314 FZ -16.4
495 498 501 504 FY -135 MZ 101.25
UNIT METER MTON
LOAD 2 LOADTYPE Live TITLE LL
UNIT METER KN
JOINT LOAD
505 FX -1.325 FY -369.54 FZ 0.127
502 FX -24.2 FY -378.227 FZ -2.32
499 FX -10.981 FY -297.031 FZ 113.6
496 FX -7.672 FY -317.261 FZ -111.46
UNIT METER MTON
LOAD 3 LOADTYPE Wind TITLE WL1(+X)
UNIT METER KN
JOINT LOAD
505 FX 36.044 FY 154.82 FZ 0.462
502 FX 10.232 FY 10.357 FZ 0.525
499 FX 35.211 FY 60.358 FZ -0.5
496 FX -37.045 FY 48.963 FZ 0.042
UNIT METER MTON
LOAD 4 LOADTYPE Wind TITLE WL2(-X)
UNIT METER KN
JOINT LOAD
505 FX -36.044 FY -154.82 FZ 0.462
502 FX -10.232 FY -10.357 FZ 0.525
499 FX -35.211 FY -60.358 FZ 0.5
496 FX -37.045 FY -48.963 FZ -0.042
UNIT METER MTON
LOAD 5 LOADTYPE Wind TITLE WL3(+Z)
UNIT METER KN
JOINT LOAD
505 FX -2.084 FY -105.51 FZ 20.155
502 FX 4.243 FY 103.29 FZ 19.811
499 FX -0.51 FY -55.665 FZ 47.47
496 FX 3.307 FY 61.503 FZ 47.883
UNIT METER MTON
LOAD 6 LOADTYPE Wind TITLE WL4(-Z)
UNIT METER KN
JOINT LOAD
505 FX 2.084 FY 105.51 FZ -20.155
502 FX -4.243 FY -103.29 FZ -19.811
499 FX 0.51 FY 55.665 FZ -47.47
LOAD 7 LOADTYPE None TITLE TL1_TEMP (+25)
LOAD 8 LOADTYPE None TITLE TL2_TEMP (-25)
***
UNIT METER MTON
LOAD COMB 101 1.4 DL
1 1.4
LOAD COMB 102 1.2 (DL+TL1) + 1.6 LL

1 1.2 2 1.6 7 1.2
LOAD COMB 103 1.2 (DL+TL2) + 1.6 LL
1 1.2 2 1.6 8 1.2
LOAD COMB 104 1.2 DL+ 1.6 WL(+X) + 0.5 LL
1 1.2 3 1.6 2 0.5
LOAD COMB 105 1.2 DL+ 1.6 WL(-X) + 0.5 LL
1 1.2 4 1.6 2 0.5
LOAD COMB 106 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
1 1.2 5 1.6 2 0.5
LOAD COMB 107 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
1 1.2 6 1.6 2 0.5
LOAD COMB 108 0.9 DL + 1.6 WL(+X)
1 0.9 3 1.6
LOAD COMB 109 0.9 DL + 1.6 WL(-X)
1 0.9 4 1.6
LOAD COMB 110 0.9 DL + 1.6 WL(+Z)
1 0.9 5 1.6
LOAD COMB 111 0.9 DL + 1.6 WL(-Z)
1 0.9 6 1.6
LOAD COMB 112 1.2 DL + 0.8 WL(+X)
1 1.2 3 0.8
LOAD COMB 113 1.2 DL + 0.8 WL(-X)
1 1.2 4 0.8
LOAD COMB 114 1.2 DL + 0.8 WL(+Z)
1 1.2 5 0.8
LOAD COMB 115 1.2 DL + 0.8 WL(-Z)
1 1.2 5 0.8
****
****Serviceanbility check
LOAD COMB 201 1. DL
1 1.0
LOAD COMB 202 1 (DL+TL1) + 1 LL
1 1.0 2 1.0 7 1.0
LOAD COMB 203 1 (DL+TL2) + 1 LL
1 1.0 2 1.0 8 1.0
LOAD COMB 204 1 DL+ 1 WL(+X)
1 1.0 3 1.0
LOAD COMB 205 1 DL+ 1 WL(-X)
1 1.0 4 1.0
LOAD COMB 206 1 DL+ 1 WL(+Z)
1 1.0 5 1.0
LOAD COMB 207 1 DL+ 1 WL(-Z)
1 1.0 6 1.0
LOAD COMB 208 0.6 DL + 1 WL(+X)
1 0.6 3 1.0
LOAD COMB 209 0.6 DL + 1 WL(-X)
1 0.6 4 1.0
LOAD COMB 210 0.6 DL + 1 WL(+Z)
1 0.6 5 1.0
LOAD COMB 211 0.6 DL + 1 WL(-Z)

```
1 0.6 6 1.0
LOAD COMB 212 1 DL+ 0.75 WL(+X)+ 0.75 LL
1 1.0 3 0.75 2 0.75
LOAD COMB 213 1 DL+ 0.75 WL(-X)+ 0.75 LL
1 1.0 4 0.75 2 0.75
LOAD COMB 214 1 DL+ 0.75 WL(+Z)+ 0.75 LL
1 1.0 5 0.75 2 0.75
LOAD COMB 215 1 DL+ 0.75 WL(-Z)+ 0.75 LL
1 1.0 6 0.75 2 0.75
****
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
UNIT METER KN
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
FC 28000 ALL
FYMAIN 414000 ALL
MINMAIN 16 ALL
MINSEC 10 ALL
DESIGN COLUMN ALL
END CONCRETE DESIGN
*LOAD LIST 1 TO 10
*PERFORM ANALYSIS
*LOAD LIST 101 TO 123
***
*PARAMETER 1
*CODE LRFD
*FU 41000 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*FYLD 27500 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*KX 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KY 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 261 268 -
*269 TO 292
*KZ 0.85 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*LX 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LZ 2 MEMB 262 TO 267
*LY 1.5 MEMB 81 82 175 TO 184 262 TO 267
*LY 5 MEMB 1 TO 4 12 TO 16 18 28 29 31 33 36 37 39 41 47 49 51 53 TO 55 58 -
*123 129 TO 131 201 202 209 TO 211 213 235 236 242 243 256 257 280 TO 282
*LX 3 MEMB 43 45 125 212 214
*LY 3 MEMB 43 45 125 212 214 253
*LZ 3 MEMB 81 82 175 TO 184
*LZ 4 MEMB 8 10 19 27 35 42 48 52 62 113 118 153 TO 160 169 TO 172 194 TO 197 -
*205 TO 208 229 230
*LZ 5 MEMB 1 TO 6 9 11 TO 17 24 25 28 TO 33 36 37 39 41 44 TO 47 49 TO 51 53 -
*54 TO 58 63 67 TO 80 111 116 123 129 TO 131 135 TO 140 144 TO 149 165 167 -
*185 186 201 202 209 TO 213 258 TO 261 268 TO 279 286
*LZ 5 MEMB 43 45 125 212 214
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*LZ 4.5 MEMB 17 25 46 50 111 135 TO 140 270 TO 279 286
*MAIN 200 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*RATIO 1 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TMAIN 300 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*TRACK 2 MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*PARAMETER 2
*CODE LRFD
*DJ2 40 MEMB 69 77 78
*DJ1 41 MEMB 69 77 78
*DJ2 10 MEMB 11 67 70 185
*DJ1 9 MEMB 11 67 70 185
*DJ2 7 MEMB 5 68 71 186
*DJ1 5 MEMB 5 68 71 186
*DJ2 50 MEMB 81 175 TO 179
*DJ1 48 MEMB 81 175 TO 179
*DJ2 51 MEMB 82 180 TO 184
*DJ1 49 MEMB 82 180 TO 184
*DJ2 42 MEMB 72 79 80
*DJ1 43 MEMB 72 79 80
*DFF 500 MEMB 5 6 11 67 TO 74 77 TO 82 175 TO 186 258 259
*DFF 360 MEMB 1 TO 4 12 TO 15 28 29 31 33 36 37 39 41 43 45 47 49 51 -
*53 TO 55 123 125 129 TO 131 201 202 209 TO 214
**FREQUENCY CHECK FOR TOP FLOOR DONE BY DEFLECTION VALUES
*DJ1 4 MEMB 3 15 31 39 130 201 210
*DJ2 8 MEMB 3 15 31 39 130 201 210
*DJ2 6 MEMB 4 13 33 41 202 213
*DJ2 10 MEMB 43 49 53 55 125 131 214
*DJ2 9 MEMB 45 47 51 54 212
*DJ2 5 MEMB 2 12 28 36 211
*DJ1 2 MEMB 4 13 33 41 202 213
*DJ1 29 MEMB 45 47 51 54 212
*DJ1 27 MEMB 43 49 53 55 125 131 214
*DJ1 3 MEMB 1 14 29 37 123 129 209
*DJ1 1 MEMB 2 12 28 36 211
*CHECK CODE MEMB 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 TO 292
*STEEL TAKE OFF LIST 1 TO 19 24 TO 37 39 41 TO 63 67 TO 107 110 TO 123 125 -
*126 TO 292
FINISH